FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Preterm Labour, PROM & PPROM

Tutorial | Week 1 · Foundations, ANC & early pregnancy | Obstetrics | Dr Nassimu

Preterm birth (<37 completed weeks) is the leading cause of newborn death in Uganda. Confirm the diagnosis (sterile speculum, not digital examination, when membranes may have ruptured), exclude infection, then buy time for antenatal corticosteroids, $MgSO_4$ neuroprotection and in-utero transfer, give erythromycin for PPROM, and deliver promptly if chorioamnionitis develops.

Must know before the ward round

1. Preterm birth = before 37+0 weeks; extremely preterm <28, very preterm 28–<32, moderate-to-late preterm 32–<37 weeks.
2. PROM = rupture of membranes before labour at ≥37 weeks; PPROM = before labour and before 37 weeks. Diagnose with a sterile speculum (pooling), NOT a digital VE.
3. Nitrazine turns blue with amniotic fluid (pH 7.0–7.5 vs vaginal 4.5–6.0): false positives with blood, semen, urine, BV and antiseptics; fern = arborisation on a dried slide.
4. Confirm GA (earliest scan) before steroids - fundal height alone can give steroids to a term baby.
5. WHO 2022 - antenatal corticosteroids for 24–34 weeks (Uganda usually 28–34) if preterm birth is likely within 7 days, GA is reliable, no maternal infection, and adequate childbirth and newborn care are available. Dexamethasone 6 mg IM 12-hourly × 4 doses.
6. Tocolysis only to buy 48 h for steroids/transfer - nifedipine 20 mg orally, then 10–20 mg every 6–8 h for up to 48 h. Never with chorioamnionitis, abruption or fetal distress, or at ≥34 weeks.
7. $MgSO_4$ for fetal neuroprotection if birth is expected before 32 weeks (reduces cerebral palsy) - 4 g IV over 20–30 min, then 1 g/h until birth or 24 h.
8. PPROM - erythromycin 250 mg orally 6-hourly for 10 days (or until labour). Avoid co-amoxiclav (NEC). No antibiotics for preterm labour with intact membranes unless GBS/infection.
9. Chorioamnionitis (fever ≥38°C + maternal pulse >100, FHR >160, uterine tenderness, foul liquor or WBC >15) - ampicillin + gentamicin (± metronidazole) and deliver regardless of gestation; no tocolysis.
10. In PPROM, a rising maternal pulse or FHR often precedes fever - temperature, pulse and FHR 4-hourly; remember steroids raise the WBC for about 72 h.
11. Intrapartum GBS prophylaxis (preterm labour, ROM >18 h, fever) - benzylpenicillin 3 g IV then 1.5 g 4-hourly, or ampicillin 2 g IV then 1 g 4-hourly.
12. Kangaroo mother care, started immediately after birth for babies <2.0 kg, saves lives.

1 Definition
2 Epidemiology
3 Aetiology and Risk Factors
4 Pathophysiology
5 Classification / Staging
6 Clinical Features
7 Differential Diagnoses
8 Clinical Workup
9 Management
10 Complications
11 Prognosis
12 Prevention
13 Special Populations
14 Key Points
15 References

1. Definition

Preterm birth is birth after viability and before **37 completed weeks** (before 37+0).

Term	Definition
Preterm birth	Birth after viability and before **37+0 weeks**
Threatened preterm labour	Regular painful contractions before 37 weeks **without** progressive cervical change
Established preterm labour	Regular contractions **with** progressive cervical change (effacement and dilatation; often ≥2–4 cm) before 37 weeks
PROM (prelabour rupture of membranes)	Rupture of membranes **before the onset of labour** at **≥37 weeks**
PPROM (preterm PROM)	Rupture of membranes before labour **and before 37 weeks**
Prolonged rupture of membranes	Membranes ruptured **>18 hours** before delivery (a risk factor for neonatal GBS/sepsis; UK uses >24 h)
Latency period	Time from rupture of membranes to onset of labour
Chorioamnionitis (intra-amniotic infection)	Infection of the membranes, amniotic fluid, placenta and/or fetus
Low birth weight (LBW)	<2,500 g (VLBW <1,500 g; ELBW <1,000 g); may be preterm, small for gestational age, or both
Cervical insufficiency	Painless second-trimester cervical dilatation (classically 16–24 weeks) from a structurally weak cervix; a separate mechanism from preterm labour

Gestational subgroups are in section 5.

The core idea. Most of what we do in threatened preterm labour is **for the baby**: we cannot reliably stop labour, but we can **prepare** the baby with **steroids** (lungs, brain, gut), **$MgSO_4$** (brain), **antibiotics** (infection) and **birth in the right place** (a facility with newborn care). **Infection is the enemy and the most common cause**: if there is chorioamnionitis, the baby must come out, whatever the gestation. And **when membranes might have ruptured, do a sterile speculum examination, not a digital vaginal examination**: every finger in the vagina shortens the time to labour and adds infection.

> **UGANDA CONTEXT**
> **Viability** in Uganda: the MoH and UCG define abortion as pregnancy loss **before 28 weeks**, and most Ugandan hospitals regard 28 weeks as the threshold for active obstetric intervention for the fetus. National referral hospitals and private NICUs (Mulago/Kawempe, Mengo, Nsambya) now resuscitate babies from about **26 weeks** and sometimes earlier. WHO recommends steroids from **24 weeks** only where the newborn can receive adequate care, so the gestation at which you act depends on **your facility**. Always ask your senior, "What is our threshold here?"

2. Epidemiology

- Preterm birth is the **commonest cause of death in newborns and children under 5** worldwide. Globally about **1 in 10 babies is born preterm** (around 13 million per year). Sub-Saharan Africa and South Asia carry most of the burden and most of the deaths.
- In Uganda the preterm birth rate is estimated at roughly **12–14%**, and **neonatal mortality is about 22 per 1,000 live births** (UDHS 2022). Prematurity (with birth asphyxia and sepsis) is one of the three big killers.
- **Three routes to preterm birth**: **spontaneous preterm labour** (about 45%), **PPROM** (about 30%) and **provider-initiated (iatrogenic) preterm birth** (about 25%, e.g. severe pre-eclampsia, APH).

- Survival depends dramatically on where a baby is born. A 32-week baby in a high-income country almost always survives; in low-resource settings many die from **hypothermia, hypoglycaemia, infection, feeding problems and respiratory distress**, most of which are preventable with **thermal care, KMC, breast milk, infection prevention and CPAP**.
- **Antenatal corticosteroids** are cheap (dexamethasone costs a few hundred shillings) but under-used. The WHO **ACTION-I** trial (which included sites in low-resource countries) showed dexamethasone reduces neonatal death when given to women 26–34 weeks with **accurate dating and adequate newborn care**, whereas the earlier ACT trial showed harm when steroids were given broadly in settings with poor gestational dating and weak newborn care. This is why WHO attaches conditions.

> **UGANDA CONTEXT**
> Realities in Uganda: many women do not know their LNMP and have no early scan; **fundal height alone overestimates or underestimates gestation**, so wrongly giving steroids to a term baby is a real risk. Many HC IVs lack a newborn unit, CPAP and continuous oxygen, so **in-utero transfer** (sending the mother before birth) is often the most valuable single act. KMC units now exist at most regional referral hospitals and many HC IVs.

3. Aetiology and Risk Factors

Triggers by pathway (details in section 4)

- **Infection / inflammation** (the commonest, especially <32 weeks): bacterial vaginosis, UTI/asymptomatic bacteriuria, **malaria**, STIs (gonorrhoea, chlamydia, trichomonas), chorioamnionitis, pyelonephritis, periodontal disease, HIV.
- **Maternal/fetal stress**: physical or psychological stress, pre-eclampsia, FGR, heavy manual work.
- **Decidual haemorrhage**: abruption, APH, threatened miscarriage earlier in pregnancy.
- **Uterine overdistension**: **twins/multiples**, polyhydramnios, uterine anomalies.
- **Cervical insufficiency**: previous cone biopsy/LLETZ, repeated dilatation (D&Cs), obstetric trauma, congenital.
- **Membrane weakening (PPROM)**: infection-driven collagen breakdown, smoking, bleeding, polyhydramnios, invasive procedures.

Risk factors

Non-modifiable	Modifiable
Previous preterm birth or second-trimester loss (the strongest; recurrence about 15–30%)	UTI, asymptomatic bacteriuria, malaria, BV, STIs
Multiple pregnancy, uterine anomaly	Smoking, alcohol
Cervical surgery (LLETZ, cone), repeated D&Cs	Heavy manual work, stress, IPV
Age <18	Short birth interval (<24 months), anaemia, poor nutrition
Medical disease: HIV, sickle cell, diabetes, heart disease	Pre-eclampsia (iatrogenic preterm birth)

4. Pathophysiology

Four pathways to spontaneous preterm labour

Term labour is a controlled inflammatory process. Preterm labour is the same process switched on too early by one of four triggers:

Pathway	Mechanism
1. Infection / inflammation	Bacteria ascend from the vagina, stimulating cytokines (IL-1, IL-6, TNF-α) → **prostaglandins** (contractions, cervical ripening) and **matrix metalloproteinases** (weaken the membranes → PPROM)
2. Maternal/fetal stress	Activation of the **HPA axis** → placental **CRH** → ↑ oestrogen and prostaglandins
3. Decidual haemorrhage	**Thrombin** is a powerful uterotonic and activates MMPs
4. Uterine overdistension	Stretch ↑ **gap junctions** (connexin-43) and **oxytocin receptors**

All four converge on a **functional progesterone withdrawal**, rising prostaglandins and a ripening cervix.

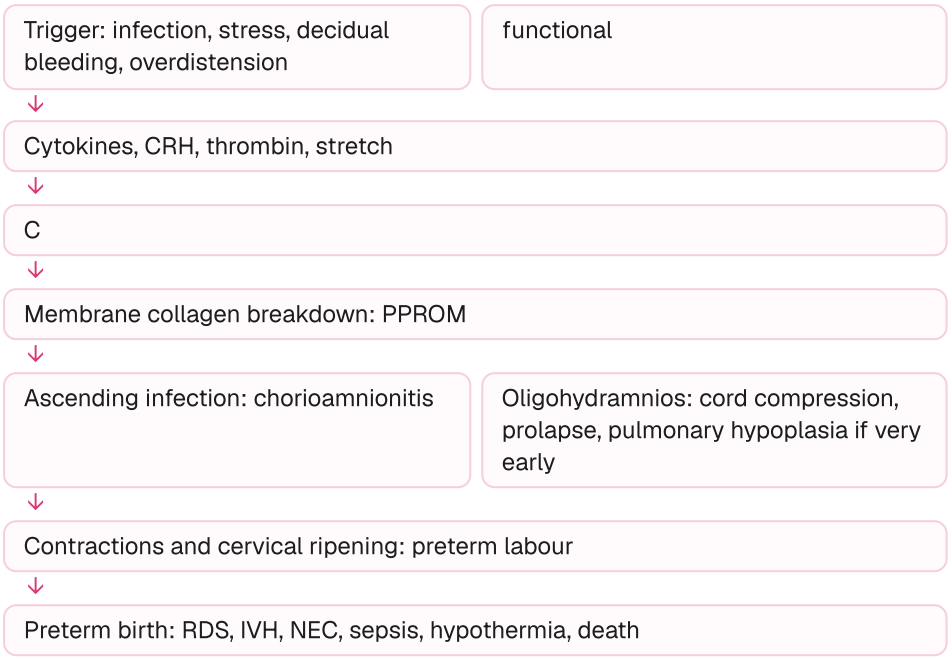
Cervical insufficiency is a separate mechanism: the cervix painlessly dilates in the **second trimester** (classically 16–24 weeks) because it is structurally weak. It is managed by **cerclage** or progesterone.

Why the membranes rupture early

The chorioamniotic membranes weaken with **collagen breakdown** (by infection-driven MMPs), apoptosis and stretching. Smoking, bleeding, polyhydramnios and invasive procedures also contribute. After rupture, **ascending infection** is the main danger. That is why we **avoid digital examinations** and give **erythromycin**.

Why prolonged pregnancy after PPROM helps and when it harms

Each week gained in utero before 34 weeks reduces prematurity complications. But the longer the latency, the higher the risk of **chorioamnionitis, abruption, cord prolapse/compression** and, if very early (<24 weeks), **pulmonary hypoplasia** and limb contractures from oligohydramnios. The art of PPROM management is balancing these.



Feature links

- Prostaglandins → regular painful contractions → progressive cervical change (established labour).
- MMPs → membrane rupture → gush then continuous trickle of clear, musty fluid → pooling on speculum, nitrazine blue, ferning, low liquor on scan.
- Ascending infection → maternal pulse rises, then fever ≥38°C → fetal tachycardia >160 → uterine tenderness → foul or purulent liquor → raised WBC.

- Infected uterus → contracts poorly → PPH after delivery.
- Decidual haemorrhage → vaginal bleeding with constant pain → abruption (tocolysis contraindicated).
- Overdistension → fundal height large for dates (twins, polyhydramnios).
- Loss of liquor → cord compression → variable decelerations, bradycardia; cord prolapse (with malpresentation).
- Very early PPROM → oligohydramnios → pulmonary hypoplasia and limb contractures.
- Weak cervix → painless mid-trimester dilatation (cervical insufficiency), not preterm labour.
- Surfactant deficiency in the preterm lung → RDS; immature germinal matrix → IVH; immature gut → NEC.

Why antenatal corticosteroids work

Dexamethasone and betamethasone cross the placenta and:

- **Mature the lungs**: induce type II pneumocytes to produce **surfactant**, thin the alveolar walls and increase lung compliance → less **respiratory distress syndrome (RDS)**.
- Mature other organs: less **intraventricular haemorrhage (IVH)**, **necrotising enterocolitis (NEC)** and neonatal death.
- Benefit starts after about **24 hours** and peaks at **48 hours to 7 days** after the first dose. **Even an incomplete course helps**, so never withhold the first dose because "she will deliver soon".
- Risks: **neonatal hypoglycaemia** (especially late preterm), maternal hyperglycaemia (monitor diabetics), and possible harm if the baby turns out to be term or if the mother is infected. That is why the WHO conditions exist.

Why $MgSO_4$ protects the fetal brain

Magnesium stabilises neuronal membranes, blocks **NMDA glutamate receptors** (reducing excitotoxic injury) and reduces inflammatory cytokines and free radicals. Given before birth under 32 weeks, it reduces **cerebral palsy by about a third** (number needed to treat about 63 below 32 weeks, fewer at earlier gestations).

Why tocolysis does not improve outcomes by itself

Tocolytics delay delivery by **48 hours to 7 days** in some women but have **never been shown on their own to reduce neonatal death or morbidity**. Their only purpose is to **buy time** for steroids to act and for transfer to a facility with newborn care.

5. Classification / Staging

By gestation

Category	Gestation
Extremely preterm	<28 weeks
Very preterm	28 to <32 weeks
Moderate preterm	32 to <34 weeks
Late preterm	34+0 to 36+6 weeks

By route: spontaneous preterm labour (about 45%), PPROM (about 30%), provider-initiated (about 25%).

By stage of labour: threatened (no cervical change) vs established (progressive cervical change).

By membrane status: intact; PPROM (<37 weeks); PROM (≥37 weeks); prolonged ROM (>18 h).

Gestation thresholds that change management: steroids **24 (Uganda usually 28) to 34 weeks**; $MgSO_4$ **<32 weeks**; tocolysis **<34 weeks**; PPROM delivery **34 or 37 weeks** (section 9).

6. Clinical Features

History

Every woman who presents before 37 weeks with pains or "water" is worked up to answer five questions, in this order: **Is she (or the baby) in danger now? Are the membranes ruptured? Is she really in labour? Is there infection? How many weeks is she, and can this facility care for a baby of that gestation?**

Ask	Why it matters (what answer changes the plan)
Dating: LNMP (certain? regular cycles? contraception?), **earliest ultrasound**, fundal height at booking	Everything (steroids, tocolysis, $MgSO_4$, place of birth) depends on accurate GA. An early scan (<24 weeks) beats LNMP; fundal height alone is unreliable
Pains: when started, how often, how long, how strong, regular? Radiating to the back?	Regular painful contractions (≥4 in 20 min or ≥8 in 60 min is one classic cut-off) suggest real labour; irregular painless tightenings = Braxton Hicks
Fluid: time of the gush (to calculate **hours since ROM**), amount (gush vs trickle), continuous or intermittent, colour (clear, bloody, meconium, green/foul), smell	PPROM vs urine/discharge. Hours since ROM → >18 h = GBS prophylaxis; meconium or foul fluid → fetal compromise/infection
Fever, chills, rigors, foul-smelling discharge, abdominal pain/tenderness	**Chorioamnionitis**: changes everything (no tocolysis, deliver)
Dysuria, frequency, loin pain	UTI/pyelonephritis, a common trigger → urine test and treat
Malaria symptoms (fever, headache, joint pains), IPTp doses, net use	Malaria is a trigger in Uganda → mRDT and treat
Vaginal bleeding (amount, painful or painless)	Abruption or placenta praevia (**no VE** until praevia is excluded)
Fetal movements	Fetal well-being, IUFD
Previous preterm birth or second-trimester loss (and whether it was painless)	The **strongest risk factor** (recurrence about 15–30%); painless = cervical insufficiency
Cervical surgery (LLETZ, cone), repeated D&Cs	Cervical insufficiency
Twins, polyhydramnios known on scan	Overdistension; changes delivery planning
Headache, visual symptoms, known high BP	Pre-eclampsia → may need iatrogenic delivery; **caution with nifedipine tocolysis on top of antihypertensive nifedipine**
Heart disease, diabetes, HIV (ART, viral load), sickle cell	Nifedipine safety, steroid glucose monitoring, eMTCT plan, neonatal plan
Vaginal discharge, STIs, sexual history	BV, gonorrhoea, chlamydia, trichomonas
Allergies (penicillin, erythromycin)	Antibiotic choice (clindamycin if penicillin-allergic for GBS)
Drugs given at the referring unit (steroid dose times, nifedipine, antibiotics)	Avoid duplicating or missing doses
Social: age <18, heavy manual work, smoking, alcohol, IPV, **distance from home, support, finances**	Risk factors; referral logistics; expectant PPROM management and KMC need a long admission

Examination

Follow the order of Hutchison's Clinical Methods (24th edition), adapted for the obstetric patient. If she is septic, bleeding or birth is imminent, the first-5-minutes assessment (section 9) comes first.

Introduction, consent and preparation

- **Introduction and identity**: introduce yourself; confirm her name, age and gestation.
- **Explanation and consent**: explain each step, especially the speculum; gain consent; **offer a chaperone**; privacy.
- **Hand hygiene**; **sterile gloves** for the speculum.
- **Position and exposure**: semi-recumbent with a left lateral tilt for the abdomen; for the speculum, let her **lie down for 20–30 minutes first** so fluid can pool.

General inspection and vital signs

- **Temperature** (≥38.0°C), **pulse** (>100 = maternal tachycardia, infection or dehydration; a rising pulse often precedes fever in chorioamnionitis), **BP** (pre-eclampsia), RR, SpO_2.
- Pallor, dehydration, signs of sepsis (RR ≥22, confusion, SBP ≤100), distress with contractions.

Hands, face and neck

- **Hands**: capillary refill, warm or cold peripheries (sepsis), pallor.
- **Face**: conjunctival pallor; flushing (fever, or nifedipine if already given).
- **Neck**: brief.
- **Renal angles**: tenderness (pyelonephritis).

Obstetric abdominal examination

Step	Finding	What it means
Inspection	Distension, scars, visible contractions	Overdistension; previous CS
Fundal height	Large for dates	Twins, polyhydramnios
Fundal height	Small for dates	FGR or wrong dates
Palpation for tone	**Uterine tenderness between contractions**	Chorioamnionitis or abruption
Contractions	Palpate for 10 minutes: frequency, duration, strength	True labour vs Braxton Hicks
Lie and presentation	Breech (common preterm), transverse	Mode of birth; cord prolapse risk after PPROM
Engagement (fifths palpable)	Fifths above the brim	Progress
Liquor volume	Reduced after PPROM	Supports PPROM
Auscultation of the fetal heart (full minute, after a contraction)	**>160 bpm**	Fetal tachycardia: early sign of chorioamnionitis
Auscultation	Bradycardia or decelerations after ROM	Cord compression or prolapse: look now

Sterile speculum examination (the key examination when ROM is suspected)

- Sterile gloves, sterile Cusco speculum, warm sterile water only (no lubricant or antiseptic in the vagina: both upset the nitrazine test).
- Look for: **pooling** of liquor in the posterior fornix; fluid from the os on **coughing (cough/Valsalva test)**: diagnostic of ruptured membranes.
- Colour (clear, blood-stained, meconium, pus) and smell of liquor.
- **Visual estimate of cervical dilatation**; **cord or fetal parts** prolapsing into the vagina.
- Take **swabs**: high vaginal and endocervical (and low vaginal + rectal for GBS where culture is available). Test fluid with **nitrazine** and make a **fern slide**.

Digital vaginal examination

- **Do it** (with care) when membranes are intact (or she is clearly in **established labour**) and there is no bleeding: dilatation, effacement, station, presentation. **Repeat after 2–4 hours** to show cervical change. This decides tocolysis and steroids.
- **Do not do it** in PPROM confirmed or suspected that is **not in labour**.
- **Do not do it** with bleeding until placenta praevia is excluded by ultrasound.

Completing the examination

- Bedside tests (section 8); point-of-care ultrasound.
- **Thank** her, cover her, **wash hands**.
- **Summarise** (model presentation in section 8).

Document the key negatives: "afebrile, pulse 84, uterus non-tender, liquor clear and non-offensive, FHR 140, no cord seen, cervix visually closed".

Red flags

RED FLAG

- **Maternal fever ≥38°C, maternal tachycardia, fetal tachycardia >160, uterine tenderness, foul-smelling liquor** → chorioamnionitis: deliver and treat
- **Vaginal bleeding with pain** → abruption (tocolysis contraindicated)
- **Cord prolapse** or cord seen/felt at the introitus → obstetric emergency (PPROM with malpresentation raises the risk)
- **Abnormal FHR** (bradycardia, decelerations) → cord compression or infection
- **Severe pre-eclampsia** → iatrogenic delivery, and no nifedipine tocolysis on top of antihypertensive nifedipine without senior input
- Signs of **sepsis**: hypotension, confusion, RR ≥22 → resuscitate and treat as maternal sepsis
- **Advanced dilatation** (≥6 cm) → delivery is imminent: prioritise $MgSO_4$, neonatal team, and the place of birth

RED FLAG

The two things that must not happen in the first 5 minutes: a **digital VE in suspected PPROM** that is not in labour, and a **digital VE with bleeding** before praevia is excluded.

Thinking like a clinician

- **Your first question is "Is this true preterm labour?"** Up to **70% of women with threatened preterm labour do not deliver within 7 days**. Overdiagnosis leads to unnecessary steroids, tocolytics and transfers. Look for **cervical change** (serial examinations 2–4 hours apart), and use transvaginal cervical length (<15 mm = high risk; >15 mm = low risk) or fetal fibronectin where available.
- **Your second question is "Why is she in preterm labour?"** Look for a trigger you can treat: **UTI, malaria, pyelonephritis, chorioamnionitis, abruption, multiple pregnancy, polyhydramnios**.
- **Your third question is "Is it safe to prolong the pregnancy?"** If there is **chorioamnionitis, abruption, fetal distress, severe pre-eclampsia, IUFD or a lethal anomaly**, the answer is no: **deliver**.
- **Your fourth question is "Where should this baby be born?"** If your facility cannot care for a 30-week baby, **transfer the mother now** (with the first steroid dose and nifedipine) rather than the baby later.
- **Painless dilatation in the mid-trimester** = cervical insufficiency, not preterm labour.
- **PPROM does not equal chorioamnionitis**, but it can become it any hour. Check temperature and pulse **4-hourly**, FHR, uterine tenderness and liquor colour/smell every shift. A rising maternal pulse or fetal heart rate

is often the **first** sign, before fever.

- **Leaking "water" may be urine** (stress incontinence is common in late pregnancy) or heavy discharge. Confirm with a speculum.
- **Steroids for the wrong baby**: if GA is uncertain and fundal height suggests 34+ weeks, steroids may harm (neonatal hypoglycaemia) without benefit. **Get an ultrasound** if possible (EFW and biometry) before giving them.
- **Common student mistakes**: doing a digital VE in PPROM; giving tocolysis to a woman with fever; forgetting the neonatal team; forgetting GBS prophylaxis in labour; giving co-amoxiclav; not calculating the GA correctly; forgetting to check steroids were actually given (look at the drug chart).

7. Differential Diagnoses

Of "contractions" before 37 weeks

Differential	Distinguishing clinical feature	Distinguishing investigation
Established preterm labour	Regular painful contractions + progressive cervical change	Serial VE (or cervical length)
Threatened preterm labour / Braxton Hicks	Irregular, often painless, no cervical change; settle with rest and hydration	Repeat VE after 2–4 h
Placental abruption	Constant pain, tender hard uterus, bleeding (may be concealed), fetal distress, hypertension	Clinical; ultrasound (retroplacental clot is not always visible)
UTI / pyelonephritis	Dysuria, loin pain, fever; uterine irritability	Urinalysis, urine culture
Malaria	Fever, rigors, headache, contractions	mRDT, blood smear
Chorioamnionitis	Fever, uterine tenderness, fetal tachycardia, foul liquor	Clinical diagnosis; WBC, CRP
Cervical insufficiency	Painless mid-trimester dilatation, bulging membranes	Speculum; cervical length
Appendicitis, red degeneration of a fibroid, ovarian torsion	Localised pain and tenderness, fever	Ultrasound, WBC
Constipation / gastroenteritis	Colicky pain, bowel symptoms	Clinical

Of "leaking fluid" in pregnancy

Differential	Distinguishing clinical feature	Distinguishing investigation
Ruptured membranes (PROM/PPROM)	Gush then continuous trickle, clear fluid, musty smell	Sterile speculum: pooling, cough test, nitrazine/fern
Urinary incontinence	Leak with coughing/laughing, smells of urine, small amounts	Speculum shows no pooling; nitrazine usually negative
Leucorrhoea / vaginal infection	Thick white or offensive discharge, itching	Wet mount, HVS
Show (mucus plug)	Blood-stained mucus, often with early labour	Speculum
Vesicovaginal fistula (rare in pregnancy)	Continuous leak, history of prior obstructed labour	Dye test

8. Clinical Workup

Bedside

Test	How / what to look for	What it changes	Available at
Sterile speculum: pooling / cough test	Liquor pooling or leaking from the os	**Confirms ROM** (the most useful test; may be negative with a small leak). No pooling after 20–30 min lying + cough = ROM unlikely (consider a pad test or repeat)	All levels with a speculum
Nitrazine paper	Vaginal pH normally **4.5–6.0** (paper stays **yellow**); amniotic fluid pH **7.0–7.5** turns it **blue**	Supports ROM. **False positive**: blood, semen, urine, BV, trichomonas, antiseptics. **False negative**: small leak, long-standing ROM with little fluid	HC III (sometimes) and above
Fern test	Swab from the posterior fornix spread on a clean slide, **air-dry for about 10 minutes**, look under low-power microscope for a **fern (arborisation)** pattern (sodium chloride crystals, oestrogen effect)	Supports ROM. **False positive**: cervical mucus, fingerprints; **false negative**: blood, too little fluid, poor drying	Where there is a microscope
Pad test	Clean pad for 1–2 h: wet, clear, musty-smelling fluid = liquor; smells of urine = urine	Useful when the speculum is equivocal	All levels
Temperature and pulse chart	**4-hourly** in PPROM; any rise	Early chorioamnionitis	All levels
FHR (Pinard/Doppler) ± CTG	Baseline >160, reduced variability, variable decelerations (cord compression after ROM)	Infection → deliver; cord compression → assess for prolapse	All levels (CTG at hospitals)
Urine dipstick	Nitrites, leucocytes (UTI); protein (pre-eclampsia); glucose	Treat UTI; recognise PE	All levels
mRDT	Positive	Treat malaria (a treatable trigger)	All levels
RBS	Baseline, especially in diabetics before steroids	Steroids raise glucose for up to 5 days	HC III and above (variable)
Hb (HemoCue)	Anaemia	Optimise before delivery; cross-match	HC III and above
Point-of-care ultrasound	Number of fetuses, presentation, viability, liquor (**single deepest pocket <2 cm** or AFI <5 cm = oligohydramnios), placental site, biometry	Presentation and praevia before VE; GA support; low liquor supports PPROM (normal liquor does not exclude it; low liquor has other causes: FGR, renal anomalies)	HC IV and above
Contraction count	Frequency over 10 min, repeated	With serial VE, confirms labour	All levels

UGANDA CONTEXT

At HC III you usually have a **speculum, nitrazine paper (sometimes), a thermometer, a Pinard/Doppler, a dipstick and an mRDT**. That is enough to diagnose PPROM or chorioamnionitis, give the first dose of dexamethasone, erythromycin or ampicillin + gentamicin, and refer. Ultrasound and FBC are usually at HC IV; cervical length, CRP, cultures and fibronectin are mostly at regional and national referral hospitals (Mengo, Mulago/Kawempe, Nsambya) or private labs.

Basic lab

Test	When	Expected / abnormal finding	How it changes management	Available at
Full blood count	Admission; **twice weekly** in expectant PPROM	WBC **>15 × 10^9/L** (a rising trend matters more than one value); anaemia	Supports chorioamnionitis. **Steroids raise WBC for about 72 h**, so interpret with the drug chart	HC IV and hospitals
Urinalysis, urine microscopy, culture	Every admission	Pyuria, bacteria, organism and sensitivities	Treat UTI/asymptomatic bacteriuria	Microscopy HC IV; culture at referral hospitals
mRDT / blood smear	Every admission (fever or not)	Parasites	Treat malaria	All levels
HIV, syphilis, HBsAg status	If unknown or due for retest	Positive	eMTCT plan, infant prophylaxis, neonatal team	All levels (rapid tests)
Blood group & cross-match	Planned delivery, bleeding, CS		Ready for APH/PPH/CS	HC IV with blood bank upward

Imaging

Test	When	Expected / abnormal finding	How it changes management	Available at
Obstetric ultrasound (biometry, EFW, presentation, liquor, placenta, number, anomalies)	Every admission; **growth and liquor every 2 weeks** in expectant PPROM	EFW and biometry consistent with dates; oligohydramnios; breech; praevia; lethal anomaly	**Confirms GA before steroids** (biometry is less accurate late in pregnancy; an early scan is best); presentation → mode of birth; EFW → place of birth	HC IV (often) and all hospitals
Transvaginal cervical length	Symptomatic, **intact membranes** only, 24–34 weeks	**<15 mm** = high risk of birth within 7 days; **>15 mm** = low risk (thresholds of 15–30 mm are used in different guidelines)	≥15 mm → avoid unnecessary steroids/tocolysis/transfer. **Never in PPROM**	Regional/national referral, private

Specialised / referral-level

Test	When	Expected / abnormal finding	How it changes management	Available at
CRP	Expectant PPROM, twice weekly where available	Rising trend	Supports infection; not diagnostic alone	Regional/national referral, private
Fetal fibronectin	Intact membranes, 24–34 weeks, no recent sex/VE/bleeding	**Negative = under 1% delivered within 7 days**	High negative predictive value; mostly used in the UK	Rarely available

Test	When	Expected / abnormal finding	How it changes management	Available at
HVS / endocervical swab (Gram stain, culture)	At speculum	Gonorrhoea, BV, trichomonas, GBS	Targeted antibiotics	Culture at referral hospitals
Low vaginal + rectal swab for GBS culture	Where culture exists	GBS carriage	Intrapartum prophylaxis (Uganda uses a **risk-based** approach instead)	Rare in Uganda
Blood cultures	Fever ≥38°C or suspected sepsis	Organism	Tailor antibiotics after delivery	Regional/national referral
CTG	Viable fetus in suspected labour or chorioamnionitis	Fetal tachycardia, reduced variability, decelerations	Fetal compromise → deliver	Referral hospitals, some HC IVs
Amniotic protein tests (IGFBP-1 / Actim PROM, PAMG-1 / AmniSure)	Equivocal ROM	Positive	Confirms ROM when the speculum is doubtful; accurate but rarely affordable	Private only

Surveillance schedule during expectant management

Situation	Maternal	Fetal
Threatened preterm labour, intact membranes	Vital signs 4-hourly; **repeat VE after 2–4 h** to look for cervical change; urine and mRDT results reviewed	FHR each assessment
PPROM, expectant	**Temperature and pulse 4-hourly**; pad colour and smell **daily**; uterine tenderness each review; **FBC ± CRP twice weekly** where available	**FHR 4-hourly**; fetal movements; **growth and liquor scan every 2 weeks**
On nifedipine tocolysis	**BP and pulse before each dose** (withhold if BP <90/50)	FHR
On $MgSO_4$ neuroprotection	RR, patellar reflexes, urine output **hourly**	Continuous or frequent FHR
After steroids in a diabetic	Blood glucose **4-hourly** for up to 5 days	

Putting it together

Model one-line summary

> "A 22-year-old G2P1 at **30+4 weeks by a 12-week scan** with **PPROM of 14 hours**, confirmed by pooling on sterile speculum, **no clinical chorioamnionitis** (T 36.9°C, pulse 86, uterus non-tender, liquor clear, FHR 142), not contracting, singleton cephalic, EFW 1.5 kg. On dexamethasone (2 of 4 doses) and erythromycin day 1."

Problem list

1. Diagnosis: PPROM / preterm labour (threatened or established) / PROM at term; hours since ROM.
2. Gestational age and how reliable it is.
3. Infection status: chorioamnionitis yes/no; triggers (UTI, malaria, BV).
4. Fetal status: FHR, presentation, EFW, liquor, number of fetuses.
5. Interventions and their times: steroids (dose number), $MgSO_4$ (<32 weeks), tocolysis (hour of 48), antibiotics (day of 10).
6. Place and timing of birth; neonatal team informed; KMC counselling.

Working diagnosis: name the syndrome, the gestation and the infection status, e.g. "PPROM at 30+4 weeks without chorioamnionitis" or "established preterm labour at 32 weeks with suspected pyelonephritis".

Model case presentation (ward round)

> "Mrs K.S., a 24-year-old G2P1 at 31+2 weeks by a 14-week scan, admitted 18 hours ago with **PPROM**, confirmed on sterile speculum by pooling of clear liquor. No contractions. She is afebrile (36.8°C), pulse 88, uterus non-tender, liquor clear and non-offensive, FHR 144. WBC 11, CRP 8. Ultrasound: singleton cephalic, EFW 1.7 kg, single deepest pocket 2.5 cm. She has had **3 of 4 doses of dexamethasone** and is on **erythromycin 250 mg 6-hourly** (day 1 of 10). **Plan**: complete steroids, continue erythromycin for 10 days, 4-hourly temperature, pulse and FHR, daily pad check, FBC/CRP twice weekly, $MgSO_4$ if she labours before 32 weeks, and GBS prophylaxis in labour. Plan delivery at 34 weeks per unit protocol, sooner if chorioamnionitis. NICU informed; KMC counselling given."

Worked example 1: is it really ruptured membranes? A woman at 33 weeks reports "water" since last night.

Finding	Result	Interpretation
Speculum after lying 30 min	No obvious pooling; small amount of fluid on **cough** from the os	Positive cough test: supports ROM
Nitrazine	Blue	Supports ROM, but she had **blood-stained discharge** (possible false positive)
Fern	Clear ferning	Supports ROM (blood would tend to give a **false negative**, so a positive fern is reassuring)
Ultrasound	Single deepest pocket 1.8 cm	Oligohydramnios supports ROM

Reading it: three independent tests agree → **PPROM**. If only the nitrazine were positive (with blood present) and there was no pooling or ferning, you would call it **"ROM not confirmed"**, do a pad test and review, rather than start a 10-day antibiotic course and steroids.

Worked example 2: is this chorioamnionitis? PPROM at 31 weeks, day 3. Steroids completed yesterday.

Finding	Result
Temperature	38.3°C (and 38.1°C 30 min later)
Maternal pulse	112/min
FHR	170 bpm baseline
Uterus	Mildly tender between contractions
Liquor	Slightly offensive
WBC	18×10^9/L

Reading it: fever ≥38°C plus maternal tachycardia, fetal tachycardia, uterine tenderness and offensive liquor = clinical chorioamnionitis by any definition. The WBC of 18 alone would be hard to interpret after steroids (they raise WBC for about 72 h), which is why the **clinical signs decide**. **Action:** ampicillin 2 g IV 6-hourly + gentamicin 5 mg/kg IV daily (+ metronidazole if CS), paracetamol, stop any tocolysis, **$MgSO_4$ neuroprotection** (<32 weeks), **deliver** (induce with oxytocin; CS only for obstetric indications), neonatal team for sepsis screen and antibiotics, watch for PPH.

> **NOTE**
> Definitions of chorioamnionitis vary. These notes use **fever ≥38°C plus at least one of** maternal tachycardia >100, fetal tachycardia >160, uterine tenderness, foul-smelling liquor, or WBC >15 × 10^9/L. The older Gibbs criteria need fever plus **two** of these; the US "Triple I" definition uses fever plus fetal tachycardia, WBC or purulent fluid. In a viva, give the criteria your unit uses and say that the clinical picture, not one number, makes the diagnosis.

Worked example 3: should she get steroids?

- LNMP uncertain, fundal height 35 cm, no scan, contracting 3 in 10 min, cervix 4 cm → **scan first** if at all possible (biometry, EFW). If EFW is about 2.6 kg and biometry suggests 35–36 weeks → **no steroids, no tocolysis**; prepare for a late preterm baby. If the scan suggests 31 weeks → dexamethasone 6 mg IM 12-hourly × 4, $MgSO_4$ if birth is expected within 24 h, consider nifedipine to allow steroids/transfer.
- 29 weeks by a 10-week scan, temperature 38.4°C, tender uterus → **no tocolysis**; antibiotics and **deliver**. Give $MgSO_4$ (<32 weeks); give the first steroid dose only if it does not delay delivery (WHO lists maternal infection as a reason not to give ACS, so follow your senior and unit protocol).

9. Management

Immediate / emergency

First 5 minutes (ABCDE plus the obstetric checks)

1. **A/B/C – vital signs**: temperature, pulse, BP, RR, SpO_2. **Fever ≥38°C, maternal pulse >100 or FHR >160** → think **chorioamnionitis/sepsis** from the start. If signs of sepsis (RR ≥22, confusion, SBP ≤100): IV access, fluids, oxygen, antibiotics within the hour.
2. **Fetal heart**: present? Rate? **Bradycardia or decelerations after membrane rupture → feel/look for a cord prolapse** (cord at the introitus is an emergency: knee-chest position, push the presenting part up, fill the bladder, call for CS).
3. **Bleeding?** If yes, **no digital VE** until placenta praevia is excluded; think abruption (tocolysis is contraindicated).
4. **Is birth imminent?** Watch her: bearing down, visible presenting part, show. If so, call the **neonatal team/most skilled newborn resuscitator**, prepare a warm room, bag-mask (preterm size), and do not transfer in active second stage.
5. **Start what cannot wait**: if GA is reasonably **24 (Uganda: usually 28) to 34 weeks** and birth is likely within 7 days with no infection → **dexamethasone 6 mg IM** now (first dose). If birth is expected within 24 h **<32 weeks** → $MgSO_4$ neuroprotection. If PPROM → erythromycin. If chorioamnionitis → ampicillin + gentamicin and plan delivery.
6. **Decide on place of birth early.** If your unit cannot look after a baby of this gestation, arrange **in-utero transfer** now (first steroid dose ± nifedipine given, escort, notes with drug times).

Definitive

Principles: what we are trying to do

1. **Confirm** the diagnosis and the **gestational age**.
2. **Exclude contraindications** to prolonging pregnancy (chorioamnionitis, abruption, fetal distress, severe PE, IUFD, lethal anomaly).
3. **Optimise the baby**: corticosteroids (24–34 weeks), $MgSO_4$ (<32 weeks), antibiotics (PPROM, GBS).
4. **Buy time if needed**: tocolysis (nifedipine) for up to 48 h to complete steroids or transfer.
5. **Decide place, timing and mode of birth**, and prepare the neonatal team.

Antenatal corticosteroids (ACS)

WHO 2022 recommendation: ACS are recommended for women with a **high likelihood of preterm birth from 24 to 34 weeks** when **all** of the following are true:

1. Gestational age can be **accurately assessed**.
2. Preterm birth is considered **imminent** (likely within **7 days**, including within 24 hours).
3. There is **no clinical evidence of maternal infection**.
4. **Adequate childbirth care** is available (recognition and safe management of preterm labour and birth).
5. The preterm newborn can receive **adequate care** if needed (resuscitation, thermal care, **KMC**, feeding support, infection treatment, safe oxygen and **CPAP**).

This applies to spontaneous preterm labour, **PPROM**, APH, hypertensive disorders, FGR, multiple pregnancy and diabetes (with glucose monitoring). WHO does **not** currently recommend routine ACS **after 34 weeks** (small respiratory benefit, more neonatal hypoglycaemia, no mortality benefit), outside research or specific senior decisions (e.g. elective caesarean at 34–36 weeks in some high-resource guidelines).

- **Give the first dose even if birth seems imminent.** A partial course still gives some benefit.
- **Repeat course**: WHO allows **one** repeat course if preterm birth did not occur within **7 days** of the first course and the woman is again at high risk of birth in the next 7 days (still <34 weeks).
- **Diabetes**: expect hyperglycaemia for up to 5 days; monitor glucose 4-hourly and adjust insulin.
- **HIV**: not a contraindication.
- **Chorioamnionitis**: do **not delay delivery** to give steroids.

> **UGANDA CONTEXT**
> Uganda practice follows WHO: **dexamethasone 6 mg IM 12-hourly × 4**. Many Ugandan units restrict ACS to **28–34 weeks** (the local threshold of viability and newborn care), extending down to about 26 weeks at centres with a NICU (e.g. Mengo, Kawempe/Mulago, Nsambya). If the gestation is uncertain, get an **ultrasound** first; do not give steroids on fundal height alone when it measures ≥34 cm.

Tocolysis

Purpose: to delay birth by **48 hours** to complete ACS and/or to **transfer** the mother in utero. It does not by itself improve newborn outcomes.

WHO 2022 makes a conditional recommendation for **nifedipine** as the tocolytic for women 24–34 weeks with a high likelihood of preterm birth (with the same conditions as for ACS). Other agents are **not** recommended: **betamimetics** (salbutamol, terbutaline, ritodrine: cause maternal tachycardia, pulmonary oedema and hyperglycaemia, and are more dangerous than they are useful) and **$MgSO_4$ for tocolysis** (ineffective as a tocolytic). **Maintenance tocolysis** beyond 48 hours has no proven benefit.

Contraindications to tocolysis

Maternal	Fetal
Chorioamnionitis / maternal infection or fever	**Fetal distress** / abnormal FHR
Antepartum haemorrhage (abruption; significant bleeding)	**Intrauterine fetal death**
Severe pre-eclampsia/eclampsia	**Lethal congenital anomaly**
Maternal cardiac disease, hypotension	GA **≥34 weeks** (the benefit is gone)
Advanced labour (cervix ≥4–6 cm per local protocol)	Severe FGR (deliver instead)

Maternal	Fetal
PPROM is a relative contraindication (most guidelines do not tocolyse; may be used briefly for steroids/transfer if no infection)	

$MgSO_4$ for fetal neuroprotection

- **Indication**: women at risk of imminent preterm birth **before 32 weeks** (WHO); NICE: offer from 24+0 to 29+6 weeks, consider from 30+0 to 33+6 weeks. Give when birth is expected within **24 hours** (established labour or planned delivery), ideally **within 4 hours** before birth.
- **Regimen**: **4 g IV over 20–30 minutes**, then **1 g/hour IV infusion until birth or for a maximum of 24 hours**.
- **Where infusion pumps are not available**, units use their eclampsia protocol (e.g. Pritchard loading). Follow local protocol and seniors.
- **Monitor** exactly as for eclampsia: RR ≥16, reflexes present, urine ≥30 mL/h; **calcium gluconate** at the bedside.
- **Do not delay urgent delivery** (e.g. for fetal distress) to give $MgSO_4$.
- Tell the neonatal team (the baby may be hypotonic).

Antibiotics

Situation	Antibiotic	Dose	Duration
PPROM (no chorioamnionitis)	**Erythromycin**	**250 mg orally every 6 hours**	**10 days** or until established labour (whichever is sooner)
PPROM (erythromycin unavailable or not tolerated)	Oral amoxicillin or a penicillin, per local protocol (**not** co-amoxiclav)	e.g. amoxicillin 500 mg every 8 hours	7–10 days
Preterm labour with intact membranes and no infection	**No routine antibiotics**		ORACLE II showed no benefit and a possible increase in cerebral palsy
GBS prophylaxis in labour (preterm labour, known GBS carrier, previous baby with GBS, maternal fever, ROM >18 h)	**Benzylpenicillin**	**3 g (5 MU) IV loading**, then **1.5 g (2.5–3 MU) IV every 4 hours**	Until delivery
	or **Ampicillin**	**2 g IV loading**, then **1 g IV every 4 hours**	Until delivery
	Penicillin allergy: **Clindamycin**	**900 mg IV every 8 hours**	Until delivery
Chorioamnionitis	**Ampicillin 2 g IV every 6 hours** + **gentamicin 5 mg/kg IV once daily** (+ **metronidazole 500 mg IV every 8 hours** if caesarean, or per UCG)		Deliver. After **vaginal birth**, stop antibiotics (or continue per UCG if still febrile). After **caesarean**, continue until **48 hours fever-free**

DRUG DOSING

Avoid co-amoxiclav (amoxicillin-clavulanate) in PPROM. The ORACLE I trial found an increased risk of **necrotising enterocolitis** in the babies. Erythromycin prolonged pregnancy and reduced neonatal morbidity.

> **UGANDA CONTEXT**
> Universal GBS screening is **not** done in Uganda. Use a **risk-based approach**: give intrapartum penicillin/ampicillin to women in preterm labour, with membranes ruptured >18 hours, with intrapartum fever, or with a previous baby affected by GBS.

Threatened / established preterm labour with intact membranes: step by step

1. **Assess** mother (vital signs, infection, bleeding) and fetus (FHR, ultrasound for GA, presentation, EFW).
2. **Confirm labour**: VE (if no bleeding) and repeat in 2–4 h. Look for and treat a **cause** (UTI, malaria).
3. If **not in labour** (no cervical change): reassure, treat triggers, observe 6–12 h, discharge with advice on danger signs. No steroids unless high likelihood of birth within 7 days.
4. If **in preterm labour, 24 (28) to 34 weeks, no contraindications**:
   - **Dexamethasone 6 mg IM** now, then 12-hourly × 4.
   - **Nifedipine** tocolysis for up to 48 h if the cervix is not advanced and steroids or transfer are needed.
   - **$MgSO_4$ neuroprotection** if <32 weeks and birth is expected within 24 hours.
   - **In-utero transfer** to a facility with a newborn unit if needed (send with a referral note, steroid and tocolysis times, and an escort).
5. If **34–36+6 weeks**: no tocolysis and generally no steroids (per WHO); allow labour to progress; prepare the neonatal team; GBS prophylaxis.
6. **In labour**: GBS prophylaxis; continuous or frequent FHR monitoring; **no routine episiotomy or forceps "to protect the head"**; vacuum is avoided **<34 weeks** (risk of intracranial haemorrhage); breech presentation is common, so discuss mode.
7. **Birth**: experienced attendant and neonatal team present; **delayed cord clamping** (at least 30–60 seconds, if the baby does not need resuscitation) reduces IVH and the need for transfusion; dry, wrap, **skin-to-skin**, warm room (25–28°C), early breastfeeding.

PPROM: step by step

1. **Confirm** with a sterile speculum (pooling, cough test ± nitrazine/fern). **No digital VE** unless in labour.
2. **Assess for chorioamnionitis**: temperature, pulse, uterine tenderness, liquor smell/colour, FHR, WBC/CRP.
3. **Ultrasound**: GA, presentation, liquor, EFW.
4. If **chorioamnionitis, abruption, fetal distress, cord prolapse or established labour → deliver** (with antibiotics, $MgSO_4$ if <32 weeks, and steroids if time allows, but never delay for them).
5. If **no complications**:
   - **Erythromycin 250 mg orally 6-hourly for 10 days.**
   - **Dexamethasone 6 mg IM 12-hourly × 4** if 24 (28) to 34 weeks.
   - **No tocolysis** routinely.
   - **Expectant management** in hospital: temperature, pulse and FHR **4-hourly**; check pads for colour and smell daily; FBC/CRP twice weekly (where available); fetal movements; growth scans every 2 weeks.
   - **Plan timing of delivery** (below).
   - **$MgSO_4$** if <32 weeks when labour starts or delivery is planned.
   - **GBS prophylaxis** once in labour.
6. **Mode of birth**: vaginal unless usual obstetric indications (e.g. breech with <32 weeks may favour CS, per senior discussion).

Timing of delivery in PPROM

Gestation	Plan
<24 weeks (and in Uganda <26–28 weeks at most facilities)	Very poor prognosis (pulmonary hypoplasia, infection, extreme prematurity). Counsel; most women opt for delivery. Individualise with a senior
24 (28) to 33+6 weeks	Expectant management with erythromycin and steroids; deliver if infection, abruption, fetal distress or labour
34+0 to 36+6 weeks	**RCOG (PPROMT trial)**: expectant management until **37+0 weeks** if no infection, as early birth increases neonatal respiratory problems and caesarean rates. **ACOG**: delivery at **34 weeks** is recommended
≥37 weeks (PROM)	**Induce labour** (oxytocin, or misoprostol if cervix unfavourable) rather than prolonged waiting; many units wait up to 24 hours for spontaneous labour at most

UGANDA CONTEXT

In many Ugandan units, women with PPROM are **delivered at about 34 weeks** rather than waiting to 37 weeks. The reasons are the difficulty of close infection surveillance on crowded wards, limited laboratory monitoring, and the relatively good survival of 34-week babies with KMC. RCOG advises waiting until 37 weeks only when surveillance is good. Know your unit's policy and explain both positions in a viva.

PROM at term (≥37 weeks)

1. Confirm by sterile speculum. Avoid VE until in labour or induction is planned.
2. Check for chorioamnionitis, meconium, presentation and FHR.
3. **Induce labour** promptly (WHO recommends induction; the TERMPROM trial showed less maternal infection and neonatal admission with induction): **oxytocin** infusion if the cervix is favourable; **misoprostol 25 µg** (vaginal every 6 h or oral every 2 h) if unfavourable and no previous CS.
4. **Antibiotics**: GBS prophylaxis if ROM >18 hours, fever, or known GBS. Routine antibiotics for all PROM at term are not recommended by WHO (UCG may advise antibiotics if ROM >18–24 h; follow local protocol).
5. Deliver immediately (with antibiotics) if chorioamnionitis.

Chorioamnionitis: step by step

Diagnosis (clinical): **maternal fever ≥38°C** plus ≥1 of: **maternal tachycardia >100**, **fetal tachycardia >160**, **uterine tenderness**, **foul-smelling or purulent liquor**, maternal WBC >15 × 10^9/L.

1. **Resuscitate** if septic (IV fluids, oxygen; screen with qSOFA: RR ≥22, altered mentation, SBP ≤100).
2. **Antibiotics immediately**: **ampicillin 2 g IV every 6 hours + gentamicin 5 mg/kg IV once daily** (+ **metronidazole 500 mg IV every 8 hours**, especially if caesarean).
3. **Paracetamol 1 g** orally/IV for fever (maternal fever worsens fetal hypoxia).
4. **Deliver**, regardless of gestation: **induce/augment** with oxytocin (vaginal preferred); caesarean only for obstetric indications. **No tocolysis.** Do not delay delivery for steroids.
5. Inform the **neonatal team**: the baby needs a sepsis screen and antibiotics.
6. Watch for **PPH** (an infected uterus contracts poorly) and **sepsis**.
7. Continue antibiotics after CS until 48 h afebrile; after vaginal birth, review.

Drug table

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
Dexamethasone (first choice in Uganda; cheap and available)	**6 mg**	**IM**	**Every 12 hours**	**4 doses** (total 24 mg); one repeat course if ≥7 days later and still <34 weeks	Neonatal hypoglycaemia; maternal hyperglycaemia (glucose 4-hourly for 5 days in diabetics); raises WBC for about 72 h; not with maternal infection (do not delay delivery)
Betamethasone (where available)	**12 mg**	IM	**Every 24 hours**	**2 doses** (total 24 mg)	As dexamethasone
Nifedipine (immediate-release)	**20 mg loading**, then **10–20 mg**	Oral	Every 6–8 hours	Up to **48 hours**	Check BP and pulse before each dose; withhold if BP <90/50. Keep total daily dose modest (side effects rise sharply above about 60 mg/day): headache, flushing, hypotension, tachycardia. **Do not combine** with $MgSO_4$ tocolysis; caution if already on antihypertensive nifedipine
Indomethacin (specialist use, <32 weeks)	50–100 mg loading, then 25 mg	Oral/rectal	6-hourly	Max 48 h	**Premature closure of ductus arteriosus** and oligohydramnios; avoid after 32 weeks
Atosiban (oxytocin antagonist)	IV regimen	IV	Infusion	48 h	Few side effects but expensive; not available in Ugandan public facilities
Magnesium sulfate (neuroprotection)	**4 g** over 20–30 min, then **1 g/h**	IV	Infusion	Until birth or max **24 h**	RR ≥16, reflexes present, urine ≥30 mL/h; calcium gluconate at the bedside; where no pump, eclampsia protocol per local guidance
Erythromycin	**250 mg**	Oral	6-hourly	**10 days** or until established labour	First choice in PPROM
Amoxicillin (PPROM alternative)	500 mg	Oral	8-hourly	7–10 days	**Not co-amoxiclav** (NEC)
Benzylpenicillin (GBS)	**3 g (5 MU)** loading, then **1.5 g (2.5–3 MU)**	IV	4-hourly	Until delivery	Risk-based use in Uganda
Ampicillin (GBS)	**2 g** loading, then **1 g**	IV	4-hourly	Until delivery	
Clindamycin (GBS, penicillin allergy)	**900 mg**	IV	8-hourly	Until delivery	
Ampicillin (chorioamnionitis)	**2 g**	IV	6-hourly	Until delivery; after CS until 48 h fever-free	

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
Gentamicin (chorioamnionitis)	**5 mg/kg**	IV	Once daily	As above	Renal function
Metronidazole (chorioamnionitis)	**500 mg**	IV	8-hourly	As above	Add if caesarean, or per UCG
Paracetamol	**1 g**	Oral/IV	6-hourly	While febrile	Maternal fever worsens fetal hypoxia
Misoprostol (term PROM, unfavourable cervix)	**25 µg**	Vaginal / oral	Vaginal 6-hourly; oral 2-hourly	Until in labour	**Not with a previous CS**
Oxytocin (term PROM, chorioamnionitis)	Per local induction protocol	IV infusion	Titrated	Until delivery	Count contractions
Vaginal progesterone (prevention)	**200 mg**	Vaginal	At night	From diagnosis to **34 weeks**	Short cervix ≤25 mm and/or previous preterm birth

Referral criteria

Level	Actions	Refer up when
HC II/III	Identify, check vital signs and FHR, do a sterile speculum if trained; **give the first dose of dexamethasone 6 mg IM** (if GA reasonably 28–34 weeks), **nifedipine 20 mg** if contracting without contraindications, **erythromycin** if PPROM, **ampicillin + gentamicin** if chorioamnionitis	Every suspected preterm labour, PPROM or chorioamnionitis: **refer in utero** with notes and an escort
HC IV / General Hospital	Full ACS course, nifedipine, $MgSO_4$, antibiotics, induction/CS, KMC, basic newborn care (oxygen)	Very preterm (<32 weeks) babies/mothers → a centre with NICU/CPAP where possible
Regional Referral Hospital	NICU (CPAP, surfactant at some), $MgSO_4$ infusions, expectant management of early PPROM, ultrasound and Doppler, KMC unit	Extremely preterm or complex cases beyond the unit's NICU capacity → NRH
National Referral Hospital (Mulago/Kawempe) and large PNFP hospitals (Mengo, Nsambya)	NICU from about 26 weeks, surfactant, full investigations	n/a

Split between RRH and NRH adapted from the source notes, which grouped them together Verify *against UCG 2023 levels of care*.

In-utero transfer checklist: first steroid dose given (time noted), nifedipine if indicated, antibiotics if PPROM or chorioamnionitis, referral note with drug times, an escort, and not in active second stage.

Monitoring and follow-up

- Surveillance schedule during expectant management: section 8.
- **Check the drug chart**: were the steroids actually given, and how many doses?
- **Before discharge after threatened preterm labour**: danger signs (fever, foul discharge, pain, leaking, bleeding, reduced movements).
- **Next pregnancy**: early booking and dating scan, prevention strategies below.

Care of the preterm baby (what obstetric students must know)

Intervention	Details
Delayed cord clamping	≥30–60 seconds if the baby does not need resuscitation
Thermal care	Warm delivery room, dry and wrap, hat, **skin-to-skin**
Kangaroo mother care (KMC)	WHO 2022: **start KMC immediately after birth** (even before stabilisation) for babies <2.0 kg, **8–24 hours a day** (continuous if possible). The iKMC trial showed **~25% fewer deaths** in babies 1.0–1.8 kg. KMC includes skin-to-skin contact, exclusive breastfeeding and early discharge with follow-up
Feeding	Early **breast milk** (expressed by cup, or nasogastric tube if <32 weeks)
Respiratory support	Oxygen by nasal prongs, **bubble CPAP** for RDS (available at many Ugandan regional hospitals)
Infection	Antibiotics (ampicillin + gentamicin) if risk factors (PPROM, chorioamnionitis, maternal fever)
Other	Vitamin K, chlorhexidine cord care per protocol, hypoglycaemia checks, jaundice monitoring, ROP screening for very preterm babies

10. Complications

Disease-related

Maternal	Neonatal
Chorioamnionitis, endometritis, **puerperal sepsis**	**RDS** (surfactant deficiency), transient tachypnoea
Placental abruption (after PPROM)	**Intraventricular haemorrhage**, periventricular leukomalacia, **cerebral palsy**
Cord prolapse, malpresentation, operative delivery	**NEC**, feeding intolerance
PPH (infected, tired uterus)	**Neonatal sepsis** (GBS, *E. coli*)
Psychological distress, postnatal depression	**Hypothermia, hypoglycaemia, jaundice**
	Pulmonary hypoplasia and limb deformities (PPROM <24 weeks)
	Long-term: developmental delay, learning difficulties

Treatment-related

Complication	Cause	Action
Hypotension, headache, flushing, tachycardia	Nifedipine	BP before each dose; withhold if BP <90/50

Complication	Cause	Action
Maternal hyperglycaemia; neonatal hypoglycaemia	Steroids	Glucose monitoring; neonatal glucose checks
$MgSO_4$ toxicity (loss of reflexes, respiratory depression)	$MgSO_4$	Hourly RR, reflexes, urine; calcium gluconate
Pulmonary oedema, tachycardia, hyperglycaemia	Betamimetics	Do not use
NEC	Co-amoxiclav in PPROM	Use erythromycin
Ductus closure, oligohydramnios	Indomethacin after 32 weeks	Avoid after 32 weeks
Retinopathy of prematurity, BPD	Neonatal oxygen	Safe oxygen, ROP screening
Intracranial haemorrhage	Vacuum <34 weeks	Avoid

11. Prognosis

- **Threatened preterm labour**: up to **70% do not deliver within 7 days**.
- **Neonatal survival** depends dramatically on gestation and on where the baby is born; in Uganda NICU centres resuscitate from about 26 weeks.
- **Steroids** reduce RDS, IVH, NEC and neonatal death when the WHO conditions are met; **$MgSO_4$** reduces cerebral palsy by about a third (<32 weeks); **KMC** reduces deaths by ~25% in babies 1.0–1.8 kg.
- **PPROM <24 weeks**: very poor prognosis (pulmonary hypoplasia, infection, extreme prematurity).
- **Recurrence**: previous preterm birth carries a recurrence risk of about 15–30%.
- Long-term: developmental delay and learning difficulties in survivors of very preterm birth.

12. Prevention

Strategy	Who	Details
Vaginal progesterone	Short cervix (≤25 mm) at 16–24 weeks, and/or previous spontaneous preterm birth	**200 mg vaginally at night** from diagnosis until **34 weeks**
Cervical cerclage	History-indicated: ≥3 previous second-trimester losses or preterm births. Ultrasound-indicated: previous preterm birth/mid-trimester loss + cervix ≤25 mm before 24 weeks. Rescue: dilated cervix with intact membranes	McDonald (purse-string) at 12–14 weeks for history-indicated; remove at **36–37 weeks** or in labour
Treat infections	All	Screen and treat **asymptomatic bacteriuria**, syphilis; **IPTp-SP** monthly from 13 weeks; ITNs
Low-dose aspirin	Women at risk of pre-eclampsia	Reduces iatrogenic preterm birth
Stop smoking, nutrition, birth spacing (≥24 months), manage anaemia	All	

> **NOTE**
> 17-hydroxyprogesterone caproate (17-OHPC) injections are no longer recommended (withdrawn in the USA in 2023 after the PROLONG trial showed no benefit).

Preventing death once preterm birth happens: steroids, $MgSO_4$, in-utero transfer, delayed cord clamping, thermal care, KMC, breast milk, infection prevention and CPAP.

13. Special Populations

- **Diabetes**: steroids cause hyperglycaemia for up to 5 days; glucose 4-hourly and adjust insulin.
- **Pre-eclampsia on nifedipine**: no nifedipine tocolysis on top of antihypertensive nifedipine without senior input; severe PE is a contraindication to tocolysis.
- **Maternal cardiac disease**: tocolysis contraindicated; avoid betamimetics.
- **HIV**: not a contraindication to steroids; eMTCT plan and infant prophylaxis; neonatal team.
- **Penicillin allergy**: clindamycin 900 mg IV 8-hourly for GBS prophylaxis.
- **Previous caesarean section**: no misoprostol for term PROM induction.
- **Multiple pregnancy, polyhydramnios**: overdistension trigger; steroids apply; plan delivery (see the *Multiple pregnancy* topic).
- **Uncertain gestation**: scan before steroids; do not give steroids on fundal height alone when it measures ≥34 cm.
- **Very early PPROM (<24 weeks, or <26–28 weeks at most Ugandan facilities)**: individualise with a senior; counsel about the poor prognosis.

14. Key Points

High-yield points

- **Sterile speculum, not fingers**, when membranes may have ruptured.
- Confirm **GA** by the earliest scan before steroids.
- Steroids **24–34 weeks** (Uganda 28–34) when the WHO conditions are met; give the first dose even if birth is near.
- Tocolysis only buys **48 h**; nifedipine; never with infection, bleeding or fetal distress.
- **$MgSO_4$ under 32 weeks** for neuroprotection.
- PPROM: **erythromycin**, not co-amoxiclav.
- Chorioamnionitis: antibiotics and **deliver**, whatever the gestation.
- A rising maternal pulse or FHR often precedes fever.
- Risk-based **GBS prophylaxis** in labour.
- **In-utero transfer** and **KMC** save lives.

Numbers & doses

Item	Value
Preterm	**<37+0 weeks**; extremely <28, very 28–<32, moderate 32–<34, late 34–36+6
Viability / abortion cut-off (Uganda)	**28 weeks** (NICU centres act from about 26)
Prolonged ROM	**>18 h** (UK >24 h) → GBS prophylaxis

Item	Value
Vaginal pH / amniotic fluid pH	**4.5–6.0** / **7.0–7.5** (nitrazine blue)
Oligohydramnios	Single deepest pocket **<2 cm** or AFI **<5 cm**
Cervical length (intact membranes)	**<15 mm** high risk; **>15 mm** low risk of birth in 7 days
Fetal fibronectin negative	**<1%** birth within 7 days
Dexamethasone	**6 mg IM 12-hourly × 4** (24 mg)
Betamethasone	**12 mg IM 24-hourly × 2**
Steroid window	**24–34 weeks** (Uganda usually 28–34); benefit from **24 h**, peak **48 h–7 days**; one repeat course if ≥7 days later
Nifedipine tocolysis	**20 mg orally**, then **10–20 mg every 6–8 h** for up to **48 h**; withhold if BP <90/50 (local daily maximum varies)
$MgSO_4$ neuroprotection	**<32 weeks**: **4 g IV over 20–30 min**, then **1 g/h** until birth or max 24 h
Erythromycin (PPROM)	**250 mg orally 6-hourly × 10 days** (or until labour)
GBS prophylaxis	Benzylpenicillin **3 g IV**, then **1.5 g 4-hourly**; or ampicillin **2 g**, then **1 g 4-hourly**; clindamycin **900 mg 8-hourly** if allergic
Chorioamnionitis	Ampicillin **2 g IV 6-hourly** + gentamicin **5 mg/kg IV daily** (+ metronidazole 500 mg IV 8-hourly if CS)
Chorioamnionitis thresholds	Fever **≥38°C**; maternal pulse **>100**; FHR **>160**; WBC **>15 × 10^9/L**
Steroid effect on WBC	Raised for about **72 h**
PPROM delivery	RCOG: **37+0** if no infection; ACOG: **34**; many Ugandan units: **about 34**
Term PROM	**Induce** (oxytocin, or misoprostol 25 µg if unfavourable)
Vaginal progesterone (prevention)	**200 mg at night** to 34 weeks (short cervix ≤25 mm or previous preterm birth)
Cerclage removal	**36–37 weeks** or in labour
KMC	Babies **<2.0 kg**, start **immediately**, 8–24 h/day
Delayed cord clamping	At least **30–60 s** if the baby does not need resuscitation

Exam traps

- **"Do a VE to see how dilated she is"** in suspected PPROM. Wrong: **sterile speculum**; no digital VE unless she is in labour.
- **"Nitrazine positive, so membranes are ruptured."** Not on its own: blood, semen, urine, BV and antiseptics cause false positives. Look for **pooling** first.
- **"Give co-amoxiclav for PPROM."** Wrong: **erythromycin** (co-amoxiclav increased NEC in ORACLE I).
- **"Give antibiotics to every woman in preterm labour."** Wrong: not with intact membranes and no infection (ORACLE II); do give **intrapartum GBS prophylaxis**.
- **"Tocolyse to prevent preterm birth."** Wrong: tocolysis only buys **48 h** for steroids or transfer; it does not improve outcome alone and is contraindicated with infection, bleeding or fetal distress.
- **"Salbutamol is the tocolytic."** Wrong: betamimetics are **not recommended** (pulmonary oedema, tachycardia); use **nifedipine**.
- **"Raised WBC after steroids = chorioamnionitis."** Not necessarily: steroids raise WBC for about 72 h; use the clinical signs (fever, tachycardia, tenderness, liquor).

- **"Steroids at 36 weeks by fundal height."** Wrong: outside the window and dating is uncertain; **scan first**.

Questions seniors ask

On the ward round

1. **How do you confirm PPROM?** History, then a **sterile speculum** after 20–30 min lying: pooling or cough test; supported by nitrazine, fern and low liquor on scan. No digital VE.
2. **What are the signs of chorioamnionitis, and which comes first?** Fever ≥38°C with maternal or fetal tachycardia, uterine tenderness, foul liquor, WBC >15. A **rising maternal pulse or FHR** often comes before fever.
3. **What are the WHO conditions for steroids?** 24–34 weeks with **accurate GA**, birth likely within **7 days**, **no maternal infection**, adequate childbirth care and adequate newborn care.
4. **Why erythromycin and not co-amoxiclav?** Erythromycin prolonged pregnancy and reduced neonatal morbidity (ORACLE I); co-amoxiclav increased **NEC**.
5. **She is 30 weeks, 3 cm, contracting and afebrile. Your plan?** Dexamethasone 6 mg IM now (×4), nifedipine 20 mg for up to 48 h if no contraindication, $MgSO_4$ if birth is expected within 24 h, GBS prophylaxis in labour, neonatal team, **in-utero transfer** if no newborn unit.
6. **When would you deliver a woman with PPROM?** Immediately for chorioamnionitis, abruption, fetal distress, cord prolapse or labour; otherwise at about 34 weeks (many Ugandan units, ACOG) or 37 weeks (RCOG, if surveillance is good).
7. **How do you tell true preterm labour from threatened preterm labour? Progressive cervical change** on serial VE 2–4 hours apart (or cervical length <15 mm / positive fibronectin where available); up to 70% of threatened cases do not deliver within 7 days.
8. **Why does in-utero transfer matter?** The mother is the best incubator: babies transferred before birth do better than those transferred after, so move the mother (with steroids started) to a unit with newborn care.

Viva questions

1. **Why avoid digital VE in PPROM?** It introduces vaginal organisms into the uterus, increasing chorioamnionitis, and shortens the latency period. A sterile speculum gives the information needed.
2. **What are the WHO conditions for giving ACS?** GA 24–34 weeks and accurate; birth likely within 7 days; no maternal infection; adequate childbirth care; adequate preterm newborn care.
3. **Dexamethasone regimen?** 6 mg IM every 12 hours × 4 doses.
4. **Which antibiotic in PPROM, and which to avoid?** Erythromycin 250 mg 6-hourly × 10 days; avoid co-amoxiclav (NEC).
5. **Contraindications to tocolysis?** Chorioamnionitis, APH/abruption, fetal distress, IUFD, lethal anomaly, severe PE, maternal cardiac disease, GA ≥34 weeks, advanced labour.
6. **What is the purpose of tocolysis?** To delay birth for 48 h so that steroids can act and/or in-utero transfer can happen; it does not improve outcomes on its own.
7. **When is $MgSO_4$ given for neuroprotection?** Imminent birth before 32 weeks; 4 g IV then 1 g/h until birth or 24 h. It reduces cerebral palsy.
8. **How is chorioamnionitis diagnosed and managed?** Fever + maternal/fetal tachycardia, uterine tenderness, foul liquor; ampicillin + gentamicin (± metronidazole), deliver regardless of gestation.
9. **What is the principle of the nitrazine test, and its false positives?** Amniotic fluid is alkaline (pH 7–7.5) and turns nitrazine paper blue; false positives with blood, semen, urine, BV, antiseptics.
10. **What is KMC?** Continuous skin-to-skin contact of the baby on the mother's chest plus exclusive breastfeeding; start immediately for babies <2.0 kg; reduces mortality, hypothermia and sepsis.

Mnemonics

> **EXAM FAVOURITE**
> **"24–34–37"**: steroids from **24** to **34** weeks; preterm means before **37**.

- **Dexamethasone "6-12-4"**: **6** mg, every **12** hours, **4** doses. **Betamethasone "12-24-2"**: **12** mg, every **24** hours, **2** doses.
- **PPROM management "SAFE"**: **S**peculum not fingers; **A**ntibiotics (erythromycin); **F**etal lung steroids; **E**xpectant management with **E**ye on infection.
- **Tocolysis contraindications "BIRDS"**: **B**leeding, **I**nfection (chorioamnionitis), **R**uptured membranes (relative), **D**istressed or **D**ead fetus, **S**evere pre-eclampsia.
- **Chorioamnionitis signs "FEVER-T"**: **F**ever, **E**levated WBC, fetal/maternal **V**ital signs up (tachycardia), **E**ffluent foul-smelling, uterine **R**igidity/tenderness, **T**reat and deliver.
- **"MAG under 32, steroids under 34"**.

OSCE tips

- **Sterile speculum station**: consent, chaperone, sterile gloves, Cusco speculum with warm sterile water (no lubricant gel, which alters nitrazine), inspect for pooling, cough test, take swabs, nitrazine/fern, and **say "I would not perform a digital examination"**.
- **Counselling station (PPROM at 30 weeks)**: explain what happened, the risks (infection, prematurity, cord problems), the plan (steroids, antibiotics, monitoring), danger signs (fever, foul discharge, pain, reduced movements), and meet the NICU team, introducing KMC.
- **Drug chart station**: write dexamethasone 6 mg IM 12-hourly × 4 with times; erythromycin 250 mg PO 6-hourly × 10 days; nifedipine 20 mg PO stat then 10–20 mg 6–8-hourly for 48 h.

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SELF-TEST

Flashcards

26 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Define preterm birth and its subcategories (WHO).	Birth before **37+0 weeks**. Extremely preterm **<28**, very preterm **28–<32**, moderate-to-late preterm **32–<37** weeks (late preterm 34+0–36+6).
2	Define PROM and PPROM.	**PROM**: rupture of membranes before the onset of labour at **≥37 weeks**. **PPROM**: rupture before labour **and before 37 weeks**.
3	Why avoid digital vaginal examination in suspected PPROM?	It carries vaginal bacteria into the uterus (↑ **chorioamnionitis**) and **shortens the latency** to labour. Use a **sterile speculum** instead, unless she is in labour.
4	Principle and false positives of the nitrazine test.	Amniotic fluid is **alkaline (pH 7.0–7.5)**, vagina acidic (4.5–6.0): paper turns **blue**. False positives: **blood, semen, urine, bacterial vaginosis/trichomonas, antiseptics**.
5	What is the fern test?	Amniotic fluid dried on a slide shows **arborisation (fern pattern)** from sodium chloride crystals under oestrogen influence. False positive: cervical mucus; false negative: blood, scanty sample.
6	WHO 2022: five conditions for antenatal corticosteroids.	At **24–34 weeks**: 1) GA **accurately** assessed; 2) preterm birth likely within **7 days**; 3) **no maternal infection**; 4) **adequate childbirth care**; 5) **adequate preterm newborn care** (resuscitation, thermal care, KMC, feeding, infection treatment, oxygen/CPAP).
7	Dexamethasone and betamethasone regimens for fetal lung maturity?	**Dexamethasone 6 mg IM every 12 h × 4 doses** (first choice in Uganda). **Betamethasone 12 mg IM every 24 h × 2 doses**.
8	When does the steroid benefit start and peak?	Begins about **24 h** after the first dose, maximal from **48 h to 7 days**. A partial course still helps, so never withhold the first dose.
9	Nifedipine tocolysis regimen?	**20 mg orally loading**, then **10–20 mg every 6–8 hours** for up to **48 hours**; check BP before each dose. Purpose: buy time for steroids/transfer.
10	Contraindications to tocolysis ('BIRDS').	**B**leeding (APH/abruption), **I**nfection/chorioamnionitis, **R**uptured membranes (relative), **D**istressed or **D**ead fetus/lethal anomaly, **S**evere pre-eclampsia. Also GA ≥34 weeks, maternal cardiac disease, advanced labour.
11	$MgSO_4$ for fetal neuroprotection: indication and regimen?	Imminent birth **before 32 weeks** (WHO). **4 g IV over 20–30 min, then 1 g/h until birth or 24 h**. Reduces **cerebral palsy** by about a third.
12	Antibiotic for PPROM and duration? Which antibiotic is avoided?	**Erythromycin 250 mg orally 6-hourly for 10 days** (or until labour). **Avoid co-amoxiclav**: ↑ necrotising enterocolitis (ORACLE I).
13	Are antibiotics given for preterm labour with intact membranes?	**No**, not routinely (ORACLE II: no benefit, possible ↑ cerebral palsy). Give only for GBS prophylaxis in labour or a specific infection (UTI, chorioamnionitis).
14	Intrapartum GBS prophylaxis regimen?	**Benzylpenicillin 3 g (5 MU) IV**, then **1.5 g (2.5–3 MU) IV every 4 h** until delivery; or **ampicillin 2 g IV then 1 g 4-hourly**. Allergy: **clindamycin 900 mg IV 8-hourly**.
15	Clinical diagnosis of chorioamnionitis?	**Maternal fever ≥38°C** plus ≥1 of: maternal tachycardia >100, **fetal tachycardia >160**, uterine tenderness, foul/purulent liquor, WBC >15 × 10^9/L.

#	Question	Answer
16	Management of chorioamnionitis?	**Ampicillin 2 g IV 6-hourly + gentamicin 5 mg/kg IV daily** (+ metronidazole 500 mg IV 8-hourly if CS), paracetamol, **deliver regardless of gestation**, no tocolysis, alert neonatal team.
17	Four pathways to spontaneous preterm labour?	1. **Infection/inflammation**; 2) **maternal/fetal stress** (HPA axis, CRH); 3) **decidual haemorrhage** (thrombin); 4) **uterine overdistension** (twins, polyhydramnios).
18	Strongest risk factor for preterm birth?	A **previous spontaneous preterm birth** (or mid-trimester loss). Others: multiple pregnancy, infections (UTI, malaria, BV), short cervix, cervical surgery, APH, polyhydramnios, teenage pregnancy, low BMI, smoking.
19	Timing of delivery in uncomplicated PPROM at 34–37 weeks: RCOG vs ACOG vs common Uganda practice?	**RCOG (PPROMT)**: expectant management to **37+0** weeks. **ACOG**: deliver at **34 weeks**. **Many Ugandan units**: deliver at about **34 weeks** (limited infection surveillance).
20	WHO 2022 kangaroo mother care recommendation?	**Start KMC immediately after birth** for babies **<2.0 kg**, for **8–24 h/day** (continuous if possible), with exclusive breastfeeding. The iKMC trial showed about **25% lower mortality**.
21	Vaginal progesterone for prevention: who and what dose?	Short cervix **≤25 mm at 16–24 weeks** and/or previous spontaneous preterm birth: **200 mg vaginally at night until 34 weeks**.
22	How do you perform a sterile speculum examination for suspected ROM, and what confirms it?	Lie her down for **20–30 min** first; sterile gloves and speculum, **water only** (no gel or antiseptic, which upset nitrazine). Look for **pooling** in the posterior fornix or fluid from the os on **coughing**; note colour, smell, cord, visual dilatation; take swabs; test with nitrazine and a fern slide. **No digital VE** unless in labour.
23	False positives and false negatives of the fern test?	**Fern**: dried fluid forms an arborisation (fern) pattern. **False positive**: cervical mucus, fingerprints. **False negative**: blood, too little fluid, poor drying (air-dry about 10 min).
24	PPROM at 31 weeks, day 3, steroids finished yesterday: WBC 18, afebrile, pulse 84, FHR 140, uterus non-tender, liquor clear. Chorioamnionitis?	**Not on this evidence.** Steroids raise WBC for about **72 h**. Without fever, maternal/fetal tachycardia, uterine tenderness or foul liquor, continue expectant care with 4-hourly temperature, pulse and FHR and repeat the FBC ± CRP (trend matters).
25	Transvaginal cervical length in threatened preterm labour: thresholds, and when is it NOT done?	**<15 mm** = high risk of birth within 7 days (treat: steroids, consider tocolysis/transfer); **>15 mm** = low risk (avoid overtreatment). Only with **intact membranes**; **never in PPROM**. Thresholds of 15–30 mm are used in different guidelines.
26	Before giving steroids for threatened preterm labour, what must you check about gestational age?	That GA is **accurate**: earliest ultrasound (ideally <24 weeks) or a certain LNMP. **Fundal height alone is unreliable**. If uncertain, **scan** (biometry, EFW) first: steroids at ≥34 weeks give little benefit and cause neonatal hypoglycaemia (WHO condition: accurate GA).

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 25-year-old G2P1 at 30 weeks (dated by an 11-week scan) presents to a HC IV with a gush of clear fluid 3 hours ago. She is not contracting, temperature 36.9°C, pulse 84, FHR 140. What is the MOST appropriate next step to confirm the diagnosis?

- A Digital vaginal examination to assess the cervix
- B Discharge with a pad and review in a week
- C Immediate caesarean section
- D Sterile speculum examination for pooling of liquor
- E Amniocentesis

2. A 22-year-old at 31 weeks has PPROM confirmed. She is afebrile with no contractions, uterus non-tender, FHR 138. Which antibiotic regimen is MOST appropriate?

- A Co-amoxiclav 625 mg 8-hourly for 10 days
- B Erythromycin 250 mg orally 6-hourly for 10 days
- C No antibiotics unless she develops fever
- D Ceftriaxone 1 g IV daily for 7 days
- E Metronidazole 400 mg 8-hourly for 7 days

3. A 28-year-old at 32 weeks with PPROM for 4 days develops a temperature of 38.6°C, pulse 112, a tender uterus and offensive liquor. FHR is 172 bpm. She has completed dexamethasone. What is the MOST appropriate management?

- A Nifedipine tocolysis and IV antibiotics
- B Continue erythromycin and observe
- C Paracetamol and repeat temperature in 4 hours
- D Repeat course of dexamethasone, then deliver in 48 hours
- E IV ampicillin + gentamicin (± metronidazole) and expedite delivery

4. A 19-year-old at 29 weeks by early scan presents with regular painful contractions every 5 minutes. The cervix is 3 cm dilated and 80% effaced, membranes intact, afebrile, no bleeding. The HC IV has no newborn unit; the regional referral hospital with CPAP is 60 km away. What is the BEST plan?

- A Give dexamethasone 6 mg IM, start nifedipine, give $MgSO_4$ if birth is imminent, and transfer her in utero
- B Deliver at the HC IV and refer the baby after birth
- C Give salbutamol tocolysis and observe for 48 hours at the HC IV
- D Start erythromycin and discharge
- E Perform ARM to speed up delivery

5. A 26-year-old at 35+2 weeks is in preterm labour, cervix 5 cm. Membranes intact, afebrile. Which of the following is MOST appropriate?

- A Nifedipine tocolysis to allow steroids
- B $MgSO_4$ for fetal neuroprotection
- C Allow labour to progress with intrapartum GBS prophylaxis and inform the neonatal team
- D Dexamethasone 6 mg IM 12-hourly × 4 before allowing delivery
- E Elective caesarean section to protect the preterm head

6. A 30-year-old at 27 weeks is in established preterm labour at Mengo Hospital (NICU available). Cervix 6 cm. She received dexamethasone 1 hour ago. Delivery is expected within a few hours. What additional intervention is MOST likely to reduce the risk of cerebral palsy in the baby?

A Nifedipine 20 mg orally
B Magnesium sulphate 4 g IV over 20–30 min, then 1 g/h until birth
C Betamethasone 12 mg IM in addition
D Routine episiotomy and forceps to protect the head
E Delayed cord clamping for 5 minutes regardless of the baby's condition

7. A 23-year-old presents at 'about 8 months' with painful contractions. She has no LNMP, no scan, and the fundal height is 35 cm. The cervix is 3 cm. There is no ultrasound available tonight. What is the MOST appropriate decision about antenatal corticosteroids? Advanced

A Give dexamethasone because every preterm baby benefits
B Give a single dose of dexamethasone 12 mg
C Give betamethasone instead of dexamethasone
D Withhold steroids because gestational age cannot be accurately assessed and the fetus is likely ≥34 weeks
E Give steroids and tocolysis for 7 days

8. A 27-year-old at 30 weeks presents with contractions and moderate painful vaginal bleeding. The uterus is tender and firm, FHR 118 with decelerations. What is the MOST appropriate statement regarding tocolysis?

A Tocolysis is contraindicated because of suspected abruption and fetal compromise
B Nifedipine is indicated to delay delivery for steroids
C Indomethacin is preferred because it does not affect BP
D Salbutamol infusion is the safest option
E Tocolysis for 48 hours, then caesarean section

9. A 24-year-old at 38 weeks has clear liquor leaking for 6 hours (confirmed on speculum). No contractions, afebrile, FHR normal, cephalic, Bishop score 4, no previous caesarean. What is the MOST appropriate management?

A Expectant management for 7 days
B Erythromycin 250 mg 6-hourly for 10 days
C Dexamethasone 6 mg IM 12-hourly × 4
D Immediate caesarean section
E Induce labour, with cervical ripening using misoprostol 25 mcg

10. A 21-year-old at 32 weeks with confirmed PPROM is on erythromycin and has completed dexamethasone. On day 3, her temperature is 37.2°C, but her pulse has risen from 84 to 108 and FHR from 140 to 168 bpm. The uterus is mildly tender. What is the MOST appropriate interpretation and action? Advanced

A Normal variation; continue observation
B Side effect of erythromycin; stop it
C Early chorioamnionitis: start IV ampicillin + gentamicin and plan delivery
D Dehydration; give 2 L IV fluids and recheck tomorrow
E Anxiety; reassure and give diazepam

11. A 29-year-old with a previous spontaneous preterm birth at 30 weeks attends ANC at 18 weeks. A transvaginal scan shows a cervical length of 18 mm. What is the MOST appropriate intervention? Advanced

- A Vaginal progesterone 200 mg nightly until 34 weeks, or cervical cerclage
- B Bed rest until 34 weeks
- C Weekly 17-hydroxyprogesterone caproate injections
- D Prophylactic nifedipine from 20 weeks
- E Routine antibiotics for 2 weeks

12. A 26-year-old at 33 weeks in preterm labour has been on nifedipine for 36 hours. She now complains of a headache and dizziness; BP is 84/48 mmHg and pulse 118. What is the MOST appropriate action?

- A Increase the nifedipine dose because contractions persist
- B Add $MgSO_4$ as a second tocolytic
- C Add salbutamol
- D Withhold nifedipine, lie her in the left lateral position, give IV fluids and reassess
- E Give hydralazine for the headache

13. A 20-year-old in preterm labour at 33 weeks has membranes that ruptured 20 hours ago. GBS status is unknown. She is afebrile. Which intrapartum antibiotic regimen is MOST appropriate?

- A No antibiotics because she is afebrile
- B Benzylpenicillin 3 g IV loading, then 1.5 g IV 4-hourly until delivery
- C Oral erythromycin 250 mg once
- D Ceftriaxone 2 g IV once after delivery
- E Gentamicin 5 mg/kg IV alone

14. A 32-year-old at 22 weeks presents with painless vaginal 'pressure'. On speculum, the cervix is 4 cm dilated with membranes bulging; there are no contractions. She had two previous painless losses at 20 and 21 weeks. What is the MOST likely diagnosis?

- A Threatened miscarriage
- B Uterine fibroid
- C Preterm labour due to chorioamnionitis
- D Placenta praevia
- E Cervical insufficiency

15. A 1.4 kg baby is born at 31 weeks at a regional referral hospital. The baby cries at birth and has good tone. The mother is well and wants to be involved in her baby's care. What intervention has the strongest evidence for reducing mortality in this baby?

- A Nursing in an incubator separated from the mother for 72 hours
- B Routine prophylactic surfactant
- C Immediate kangaroo mother care with exclusive breastfeeding
- D Formula feeds from day 1
- E Routine phototherapy

Answers & explanations

> **1. D** · Sterile speculum examination for pooling of liquor
>
> **Sterile speculum examination** (pooling in the posterior fornix, cough test ± nitrazine/fern) is the diagnostic step. **Digital VE** should be avoided when PPROM is suspected and she is not in labour (infection, shorter latency). **Caesarean** is not indicated for uncomplicated PPROM. **Discharge** without diagnosis is unsafe. **Amniocentesis** is not a routine diagnostic test for rupture of membranes.

2. B · Erythromycin 250 mg orally 6-hourly for 10 days

Erythromycin 250 mg 6-hourly for 10 days prolongs pregnancy and reduces neonatal morbidity (ORACLE I; WHO, RCOG). **Co-amoxiclav** increased **NEC** and is avoided. **Withholding antibiotics** loses a proven benefit. **Ceftriaxone** and **metronidazole alone** are not the evidence-based choices for PPROM prophylaxis.

3. E · IV ampicillin + gentamicin (± metronidazole) and expedite delivery

This is **chorioamnionitis**: IV **ampicillin + gentamicin** (± metronidazole) and **deliver regardless of gestation**, usually by induction/augmentation. **Tocolysis** is contraindicated with infection. **Continuing erythromycin** undertreats a serious infection. **Delaying delivery** for more steroids risks maternal sepsis and fetal brain injury. **Paracetamol alone** delays definitive care.

4. A · Give dexamethasone 6 mg IM, start nifedipine, give $MgSO_4$ if birth is imminent, and transfer her in utero

The best 'incubator' for transfer is the uterus. Give the **first dose of dexamethasone**, **nifedipine** tocolysis to buy time, $MgSO_4$ for neuroprotection (<32 weeks) if birth is imminent, and **in-utero transfer** to a facility with newborn care. **Postnatal transfer** of a 29-week baby carries high mortality (hypothermia, hypoglycaemia). **Betamimetics** (salbutamol) are not recommended (maternal tachycardia, pulmonary oedema). **Antibiotics** are not indicated with intact membranes and no infection. **ARM** accelerates an unwanted preterm birth.

5. C · Allow labour to progress with intrapartum GBS prophylaxis and inform the neonatal team

At **≥34 weeks** tocolysis has no role, and WHO does not recommend routine steroids (no mortality benefit, more neonatal hypoglycaemia). Allow labour, give **GBS prophylaxis** (preterm labour) and prepare the **neonatal team**. **$MgSO_4$ neuroprotection** is for births <32 weeks. **Caesarean** does not protect the preterm cephalic baby and is for obstetric indications.

6. B · Magnesium sulphate 4 g IV over 20–30 min, then 1 g/h until birth

$MgSO_4$ before 32 weeks reduces cerebral palsy by about a third. **Nifedipine** at 6 cm is unlikely to stop labour and does not reduce CP. **Extra betamethasone** doubles the steroid dose without benefit. **Routine episiotomy/forceps** does not protect the preterm brain (forceps/vacuum can cause injury). **Delayed cord clamping** (30–60 s) reduces IVH, but not if the baby needs resuscitation, and 5 minutes regardless of condition is wrong.

7. D · Withhold steroids because gestational age cannot be accurately assessed and the fetus is likely ≥34 weeks

WHO conditions require **accurate gestational age**. A fundal height of 35 cm suggests about 34–36 weeks or more; steroids at or near term give little benefit and cause **neonatal hypoglycaemia** (the ACT trial showed harm when given indiscriminately). **Betamethasone** has the same issue. A **single 12 mg** dexamethasone dose is not a recommended regimen. **Seven-day tocolysis** has no evidence and is not indicated here.

8. A · Tocolysis is contraindicated because of suspected abruption and fetal compromise

Painful bleeding, a hard tender uterus and an abnormal FHR suggest **placental abruption** with fetal compromise. Tocolysis is **contraindicated** (APH, fetal distress); delivery is needed. **Nifedipine** would mask signs and delay needed delivery. **Indomethacin** increases bleeding risk and is also contraindicated. **Salbutamol** causes tachycardia and hides hypovolaemia. **Delaying 48 hours** risks fetal death.

9. E · Induce labour, with cervical ripening using misoprostol 25 mcg

At term (**PROM**), induction reduces maternal infection and neonatal admission compared with prolonged waiting. With an unfavourable cervix and no previous CS, **misoprostol 25 mcg** (vaginal every 6 h or oral every 2 h) is appropriate, followed by oxytocin if needed. **Waiting 7 days** increases infection. **Erythromycin** is for **PPROM**. **Caesarean** has no indication. **Steroids** are not given at term.

10. C · Early chorioamnionitis: start IV ampicillin + gentamicin and plan delivery

Rising **maternal and fetal tachycardia with uterine tenderness** are early signs of **chorioamnionitis**, often before fever. Start IV antibiotics and deliver. Calling it **normal** delays treatment. **Erythromycin** does not cause these signs. **Dehydration** does not explain a tender uterus and fetal tachycardia; waiting until tomorrow is dangerous. **Anxiety** is a diagnosis of exclusion.

11. A · Vaginal progesterone 200 mg nightly until 34 weeks, or cervical cerclage

Previous preterm birth plus a **short cervix (≤25 mm) before 24 weeks → vaginal progesterone** (200 mg nightly to 34 weeks) or **ultrasound-indicated cerclage**. **Bed rest** has no benefit and raises VTE risk. **17-OHPC** was withdrawn after the PROLONG trial showed no benefit. **Prophylactic tocolysis** and **routine antibiotics** do not prevent preterm birth.

12. D · Withhold nifedipine, lie her in the left lateral position, give IV fluids and reassess

This is **nifedipine-induced hypotension** (a recognised side effect). **Withhold nifedipine**, left lateral position, IV crystalloid bolus, monitor FHR. **Increasing** the dose worsens hypotension. **Salbutamol** adds tachycardia and is not recommended. **Combining $MgSO_4$ with nifedipine** for tocolysis increases the risk of hypotension and neuromuscular block. **Hydralazine** lowers BP further.

13. B · Benzylpenicillin 3 g IV loading, then 1.5 g IV 4-hourly until delivery

She has two GBS risk factors (**preterm labour** and **ROM >18 hours**), so she needs **intrapartum GBS prophylaxis**: benzylpenicillin (or ampicillin 2 g then 1 g 4-hourly) until delivery. **No antibiotics** misses the indication. A **single oral erythromycin** dose does not achieve adequate levels. **Postnatal ceftriaxone** is too late. **Gentamicin alone** does not cover GBS reliably.

14. E · Cervical insufficiency

Painless mid-trimester cervical dilatation with recurrent similar losses = **cervical insufficiency**. Management options include rescue cerclage (if no infection or contractions) and, in future pregnancies, history-indicated cerclage at 12–14 weeks. **Threatened miscarriage** has a closed os. **Preterm labour** is painful with contractions. **Placenta praevia** presents with painless bleeding, not dilatation. **Fibroids** do not cause this picture.

15. C · Immediate kangaroo mother care with exclusive breastfeeding

WHO 2022 recommends **immediate KMC** (skin-to-skin started straight after birth) for babies <2.0 kg; the iKMC trial showed about **25% lower mortality** for babies 1.0–1.8 kg. **Separation into an incubator** delays KMC. **Prophylactic surfactant** is not routine for a vigorous baby. **Formula** increases NEC and infection compared with breast milk. **Phototherapy** is only for jaundice.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Threatened preterm labour at 30 weeks

A 20-year-old primigravida at 30+4 weeks (by 12-week scan) presents to a HC IV in Mukono with lower abdominal pains for 6 hours. She reports burning on micturition for 3 days. Temperature 37.8°C, pulse 100, BP 112/70. Contractions 2 in 10 minutes, each lasting 30 seconds. FHR 148.

1. What focused history do you take?

> Character and frequency of the pains; **leaking liquor** (PPROM); **vaginal bleeding** (abruption/praevia: no VE until excluded); **fever, rigors, headache, joint pains** (malaria); **dysuria, frequency, loin pain** (UTI/pyelonephritis); foul discharge; fetal movements; previous preterm birth; HIV status; IPTp doses and net use; drug allergies; social situation and distance to the referral hospital.

2. What examinations and investigations do you perform?

> Abdominal exam: fundal height, **uterine tenderness**, contractions over 10 minutes, lie/presentation, FHR. **Renal angle tenderness**. Sterile speculum (no pooling, no bleeding), then a **careful digital VE** (membranes intact): the cervix is 2 cm, 50% effaced. Tests: **urinalysis** (nitrites, leucocytes), urine culture, **mRDT**, FBC, ultrasound (GA, EFW, presentation, liquor, placenta).

3. Urinalysis: nitrites +, leucocytes +++. mRDT negative. Repeat VE after 3 hours: 3 cm, 70% effaced, contractions 3 in 10 minutes. What is the diagnosis and management?

> **Established preterm labour at 30 weeks, likely triggered by a UTI/pyelonephritis.** 1) **Dexamethasone 6 mg IM** now and 12-hourly × 4. 2) **Nifedipine 20 mg orally**, then 10–20 mg 6–8-hourly for up to 48 h (no chorioamnionitis: uterus non-tender, liquor not involved). 3) Treat UTI: **ceftriaxone 1 g IV daily** (or per UCG) after urine culture. 4) **$MgSO_4$** 4 g IV then 1 g/h if birth becomes imminent (<32 weeks). 5) **In-utero transfer** to a hospital with a newborn unit/CPAP, with an escort, referral note and drug times. 6) Paracetamol for fever.

4. She is transferred and labour progresses to 7 cm at 30+5 weeks. What do you prepare for the birth?

> Continue $MgSO_4$ (neuroprotection) with RR/reflex/urine monitoring. **Intrapartum GBS prophylaxis** (benzylpenicillin 3 g IV then 1.5 g 4-hourly, or ampicillin). Inform the neonatal team to attend. Prepare a warm room (25–28°C), towels, hat, bag-and-mask with preterm mask, suction, oxygen, CPAP. No routine episiotomy; **avoid vacuum <34 weeks**. **Delayed cord clamping** 30–60 s if the baby is vigorous. Active management of the third stage with oxytocin 10 IU IM.

5. A 1.5 kg baby is born and cries well. What newborn care do you advise?

> Dry, **skin-to-skin immediately (KMC)**, hat, early breastfeeding (expressed breast milk by cup/NG if needed), vitamin K, check glucose and temperature, monitor for RDS (grunting, recession; CPAP if needed), antibiotics if sepsis risk factors (maternal fever/UTI in labour: discuss with paediatrician). Continue KMC 8–24 h/day. Counsel the mother on danger signs and follow-up.

6. What advice for her next pregnancy?

> Early booking and **dating scan**; screen and treat **asymptomatic bacteriuria**; IPTp-SP and ITN; consider **transvaginal cervical length** at 16–24 weeks with **vaginal progesterone** if ≤25 mm; interpregnancy interval ≥24 months (contraception now); counsel on danger signs (contractions, leaking, fever).

Take-away: Confirm true preterm labour, find and treat the trigger, then give steroids, tocolysis to buy time,

$MgSO_4$ under 32 weeks and transfer in utero to where the baby can be cared for.

Case 2: PPROM at 29 weeks

A 27-year-old G3P2 at 29 weeks by a 10-week scan presents to Mengo Hospital with a gush of fluid 12 hours ago and a continuous trickle since. No contractions. Temperature 36.8°C, pulse 86, uterus soft and non-tender, FHR 142.

1. How do you confirm the diagnosis?

> **Sterile speculum examination** after she lies down for 20–30 min: **pooling of liquor** in the posterior fornix and/or leakage on coughing. Note colour and smell. Nitrazine (turns blue) and fern tests if available. Take an HVS. Ultrasound for liquor volume (reduced), GA, EFW and presentation. **No digital VE.**

2. PPROM is confirmed. What is your initial management?

> Admit. **Erythromycin 250 mg orally 6-hourly for 10 days**. **Dexamethasone 6 mg IM 12-hourly × 4**. No tocolysis. Baseline FBC and CRP. Monitoring: **temperature, pulse and FHR 4-hourly**, daily pad check (colour/smell), fetal movements. Inform NICU and counsel on KMC. $MgSO_4$ if labour starts before 32 weeks. GBS prophylaxis when in labour.

3. Which features would make you deliver her early?

> **Chorioamnionitis** (fever, maternal/fetal tachycardia, uterine tenderness, foul liquor, rising WBC/CRP); **placental abruption** (bleeding, pain); **abnormal FHR** or cord prolapse; **established labour**; other maternal indications (e.g. severe pre-eclampsia).

4. On day 5 her temperature is 38.4°C, pulse 110, FHR 170 and the liquor is offensive. What do you do?

> **Chorioamnionitis.** Start **ampicillin 2 g IV 6-hourly + gentamicin 5 mg/kg IV daily** (+ metronidazole if CS), paracetamol, IV access and bloods (FBC, blood cultures if available). **$MgSO_4$** 4 g IV then 1 g/h (she is <32 weeks). **Induce labour** (oxytocin, or misoprostol if unfavourable), with caesarean only for obstetric indications. No tocolysis. Neonatal team present; the baby needs a sepsis screen and antibiotics.

5. If she had remained well, when would you deliver her, and why is this debated?

> **RCOG (PPROMT trial)**: expectant management to **37+0 weeks** if no infection, because early delivery increases neonatal respiratory problems and CS without reducing sepsis. **ACOG**: deliver at **34 weeks**. **Many Ugandan units deliver at about 34 weeks** because infection surveillance on busy wards is limited and 34-week babies do well with KMC. Individualise and follow unit policy.

Take-away: PPROM: speculum not fingers, erythromycin, steroids, vigilant infection surveillance, and deliver the moment chorioamnionitis appears.

Case 3: Term PROM at a HC IV

A 31-year-old G2P1 at 39 weeks presents to a HC IV in Luweero with leaking clear liquor for 20 hours. No contractions. She had one previous normal delivery. Temperature 37.0°C, pulse 88, BP 118/76, cephalic, 3/5 palpable, FHR 136.

1. What is your diagnosis and what must you rule out?

> **Term PROM** with **prolonged rupture (>18 h)**. Rule out **chorioamnionitis** (fever, tachycardia, uterine tenderness, foul liquor, fetal tachycardia), **meconium**, **cord prolapse** (speculum), malpresentation and fetal distress. Confirm by sterile speculum.

2. Speculum confirms clear liquor; cervix appears closed. Bishop score is 4. What is your plan?

> **Induce labour** now rather than waiting further (reduces maternal infection and neonatal admissions). Unfavourable cervix and no previous CS → **misoprostol 25 mcg vaginally every 6 h** (or 25 mcg orally every 2 h) per unit protocol, then oxytocin/ARM when favourable. Start **GBS prophylaxis** (ROM >18 h): benzylpenicillin 3 g IV then 1.5 g 4-hourly, or ampicillin 2 g then 1 g 4-hourly. Monitor temperature 4-hourly, FHR, contractions.

3. Six hours after the second misoprostol dose, she has 5 contractions in 10 minutes, each lasting 70 seconds, and FHR drops to 100 bpm. What is happening and what do you do?

> **Uterine tachysystole/hyperstimulation with fetal bradycardia.** Stop further misoprostol, left lateral position, oxygen, IV fluids, and consider acute tocolysis (e.g. **terbutaline 0.25 mg SC** or nifedipine per protocol). Assess for cord prolapse and progress. If FHR does not recover, **expedite delivery** (assisted vaginal birth if fully dilated, otherwise caesarean).

4. The FHR recovers and she delivers a 3.2 kg baby vaginally. What postnatal care do mother and baby need?

> Active management of the third stage (oxytocin 10 IU IM), monitor for PPH and fever. Baby: observe for **early-onset sepsis** for at least 12–24 h (temperature, breathing, feeding), since prolonged ROM is a risk factor; treat if signs appear. Early breastfeeding. Counsel both on danger signs (fever, foul lochia; baby not feeding, fast breathing, hot or cold).

Take-away: Term PROM: confirm, exclude infection, induce promptly, and give GBS prophylaxis once membranes have been ruptured for more than 18 hours.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.